BAHRIA UNIVERSITY, (Karachi Campus)

Department of Software Engineering

Assignment 3 - Spring 2025

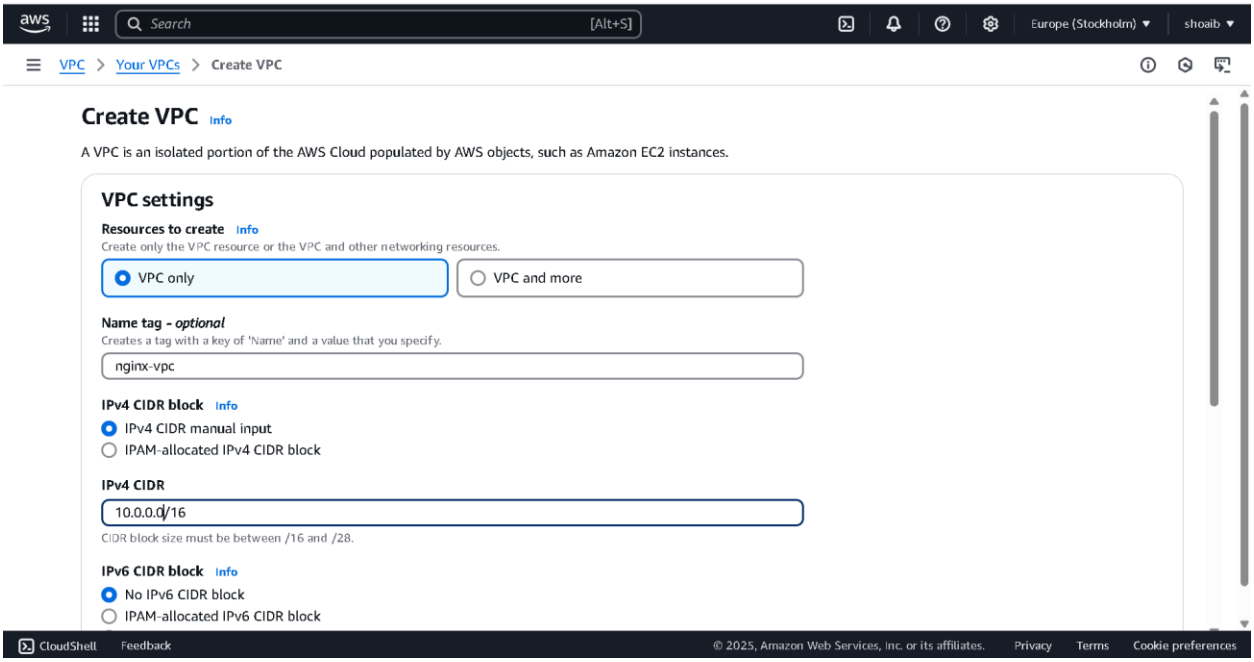
COURSE TITLE:	DevOps	COURSE CODE:	SEN-429
Class:	**BSE-8 B**	Shift:	**Morning**
Course Instructor:	Sohaib Ur Rehman	Time Allowed:	**1 Week**
Submission Date:	**18th May 2025**	Max. Marks:	**5 Marks**

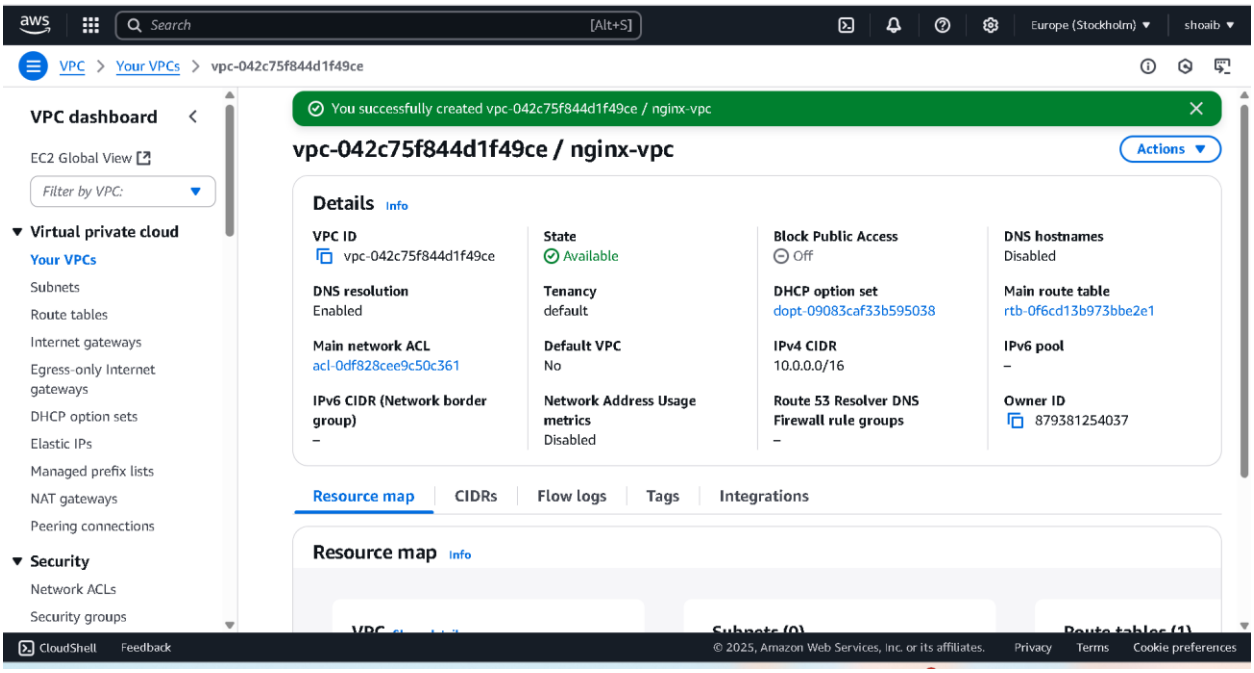
Name: Muhammad Shoaib Akhter Qadri
Enrollment No: 02-131212-009

Solution:

Here is the step by step guideline:

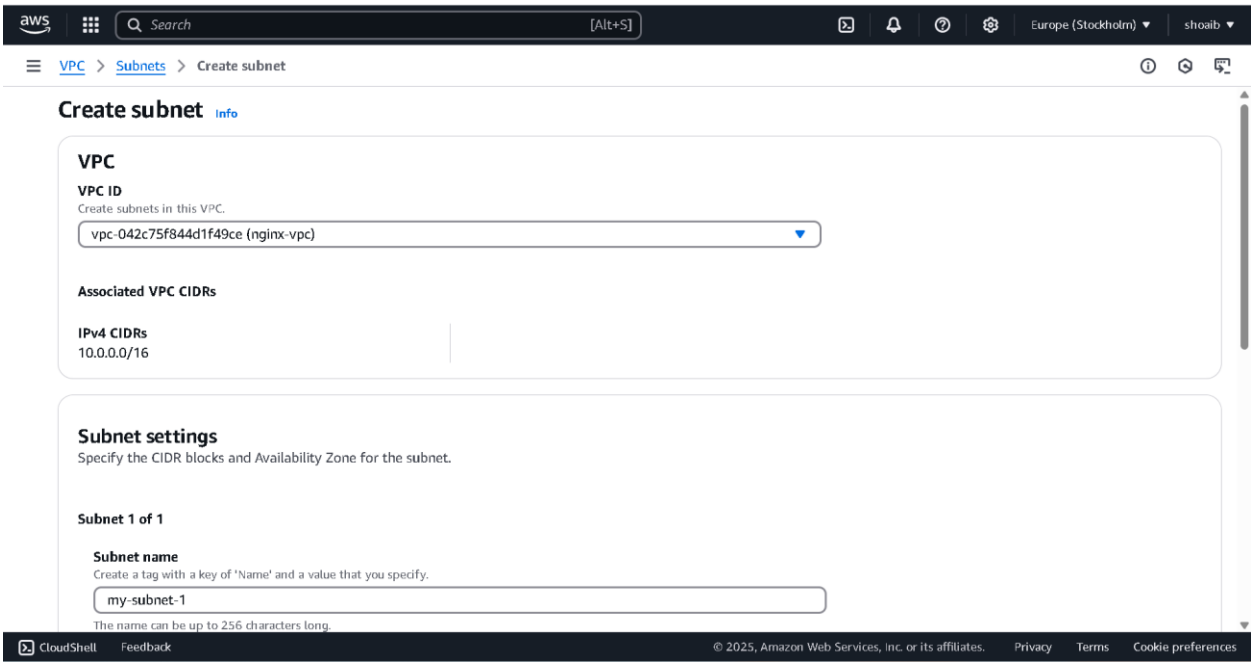
1. Create VPC First and keep nginx-vpc name and give IPv4 as 10.0.0.0/16

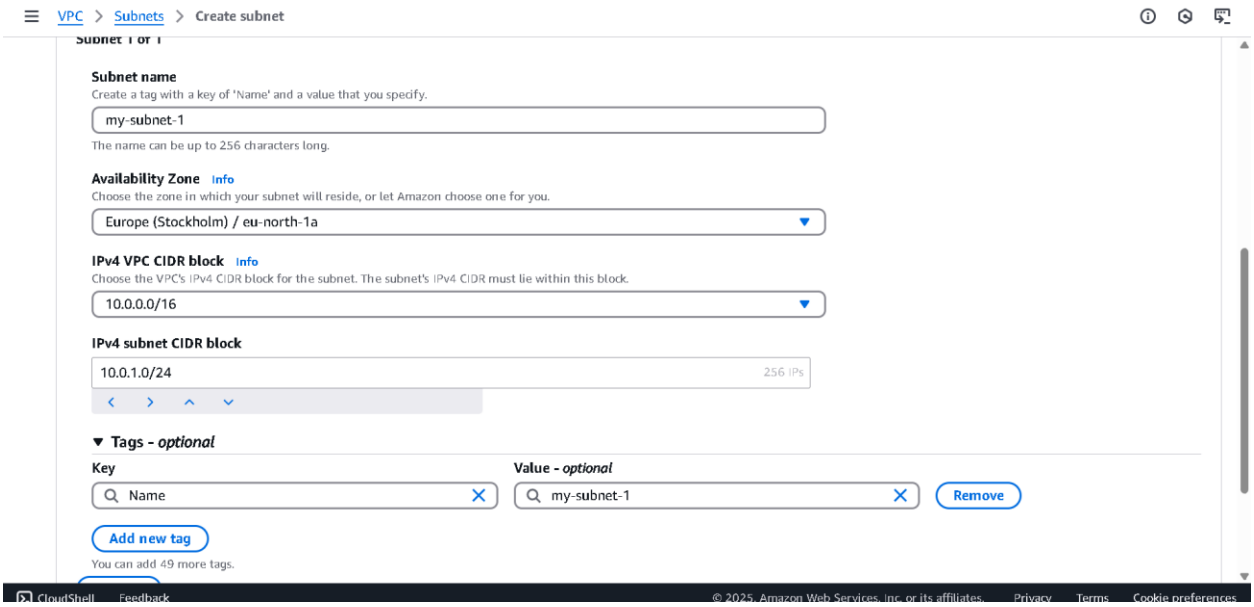




2. Create 2 Subnets under above VPC.

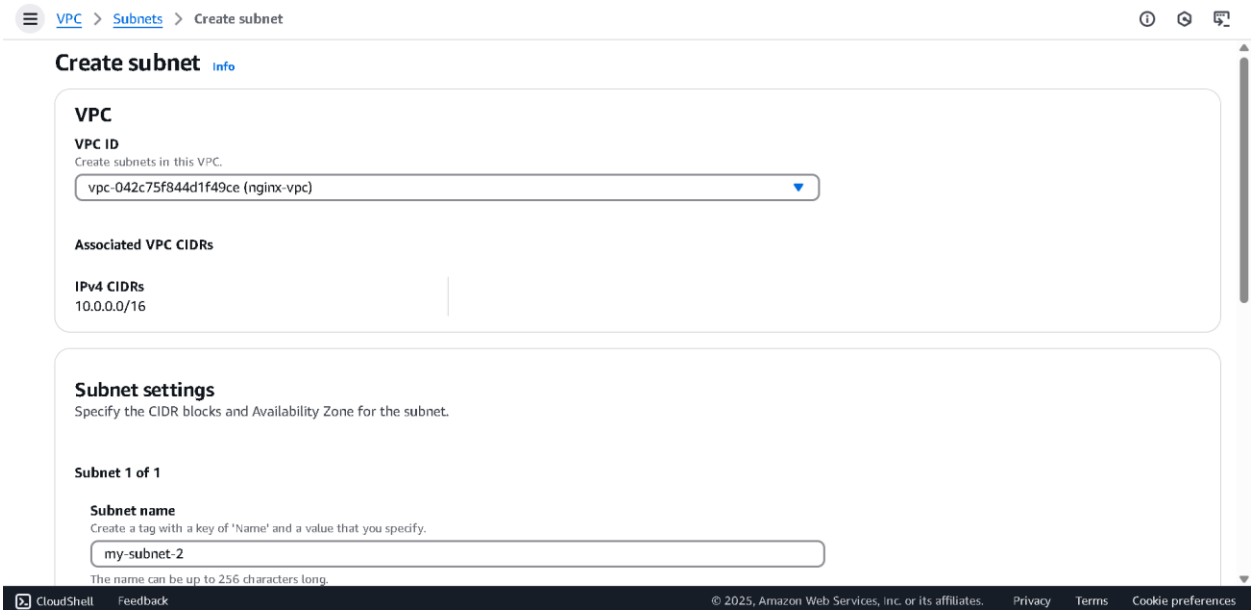
Select nginx-vpc VPC and
give subnet name as my-subnet-1 and
select availability as eu-north-1a and
select IPv4 subnet CIDR block as 10.0.1.0/24

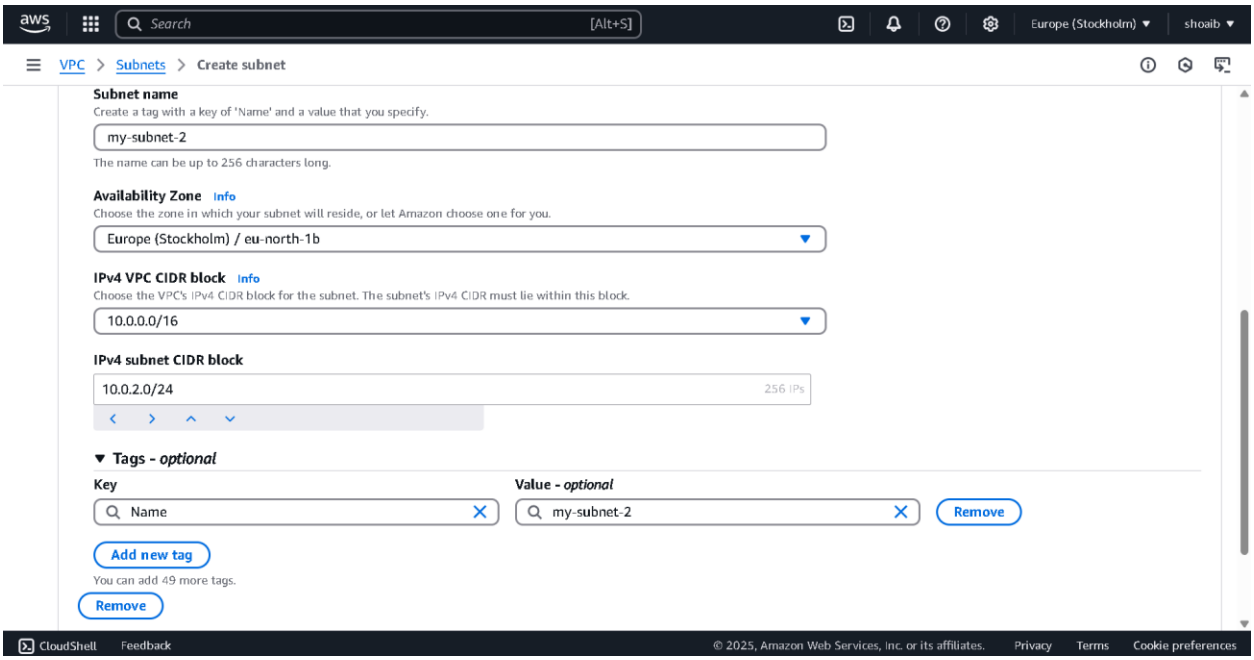




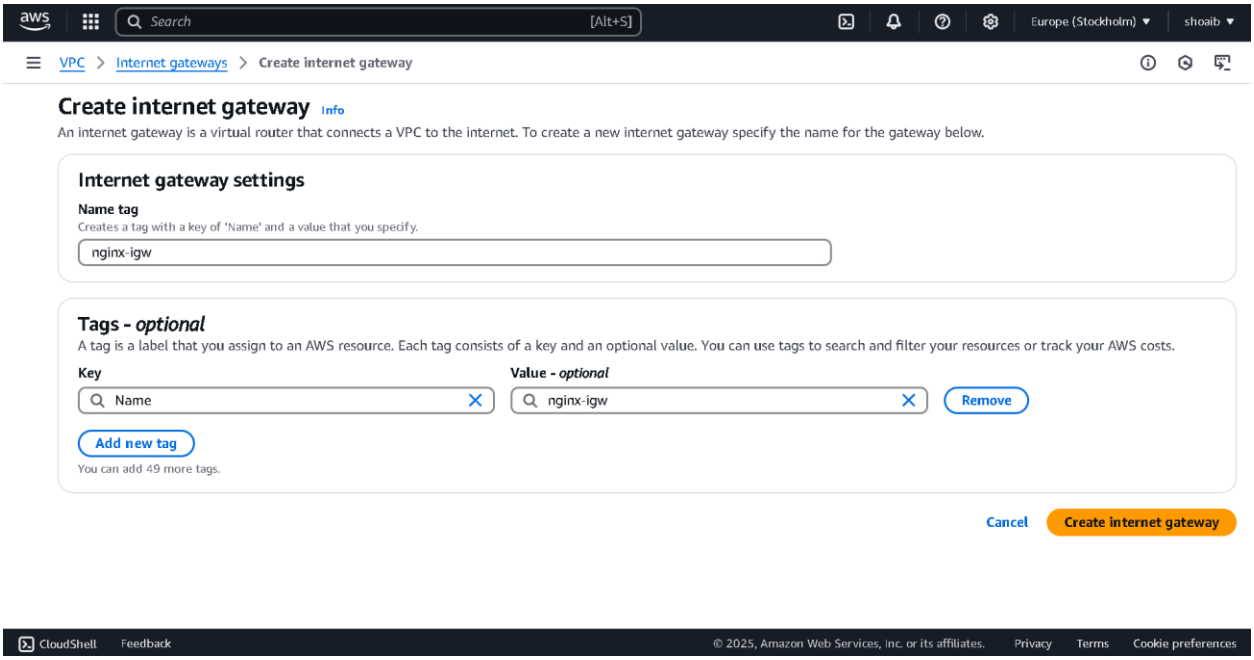
3. Now created First subnet, then create second subnet

Select nginx-vpc VPC and
give subnet name as my-subnet-2 and
select availability as eu-north-1b and
select IPv4 subnet CIDR block as 10.0.2.0/24

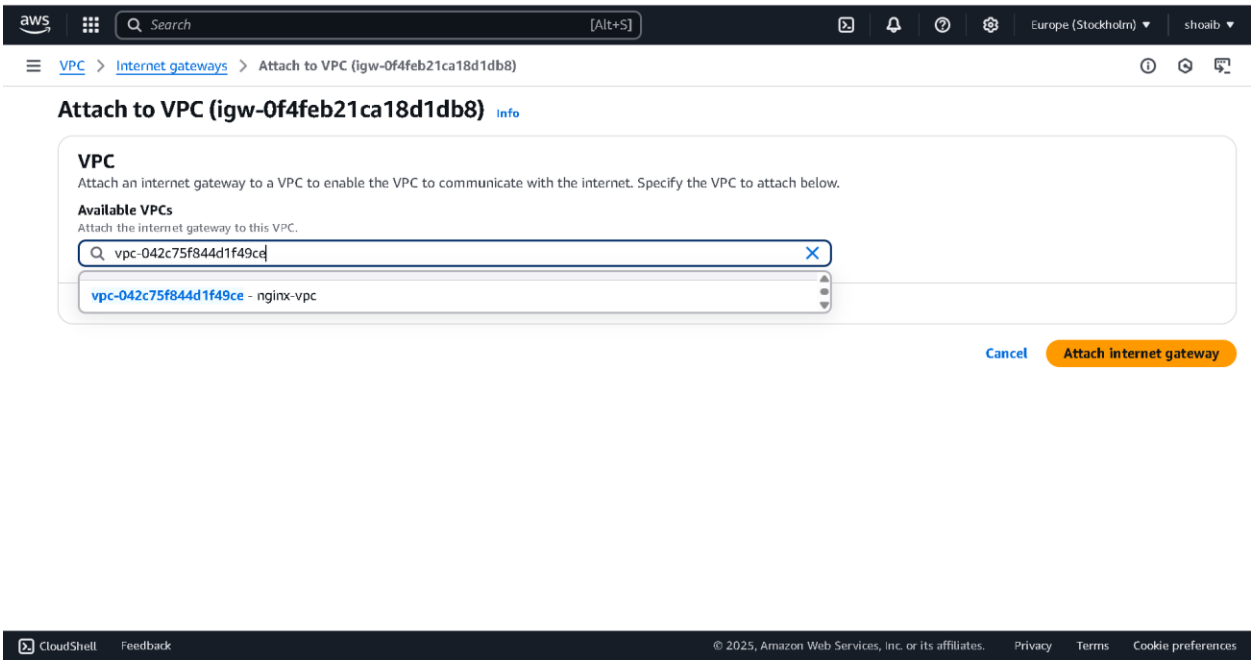




4. Attached Internet Gateway also and keep name as nginx-igw

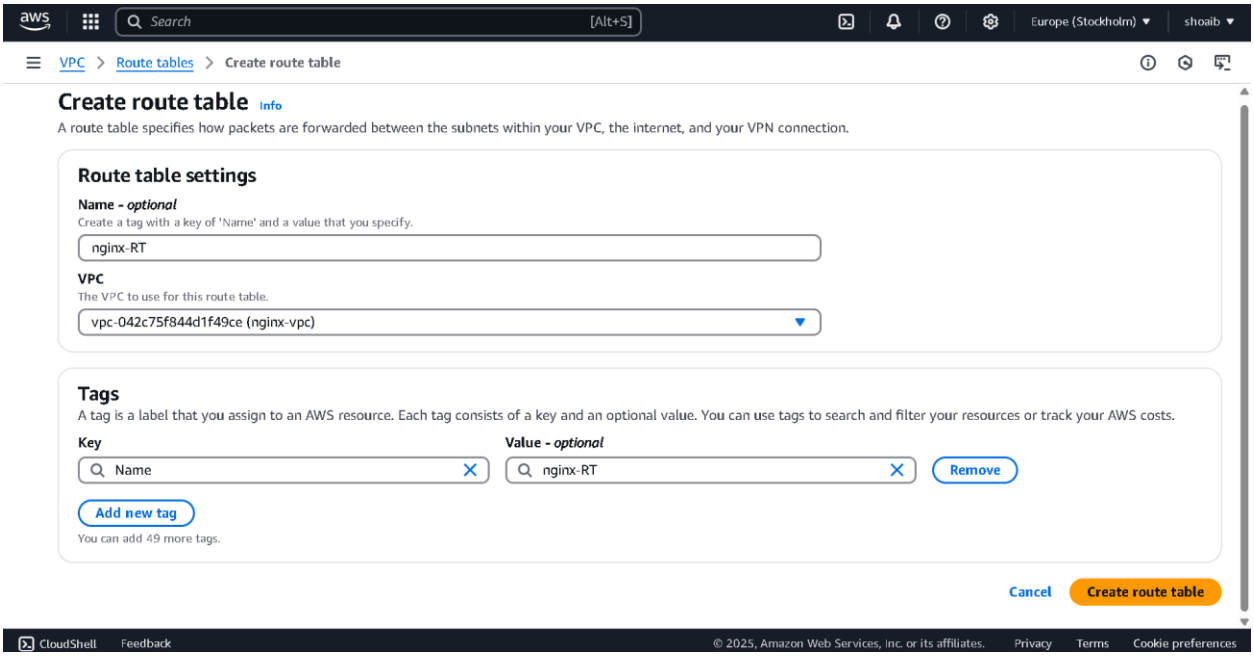


Select Internet Gateway and attached with VPC

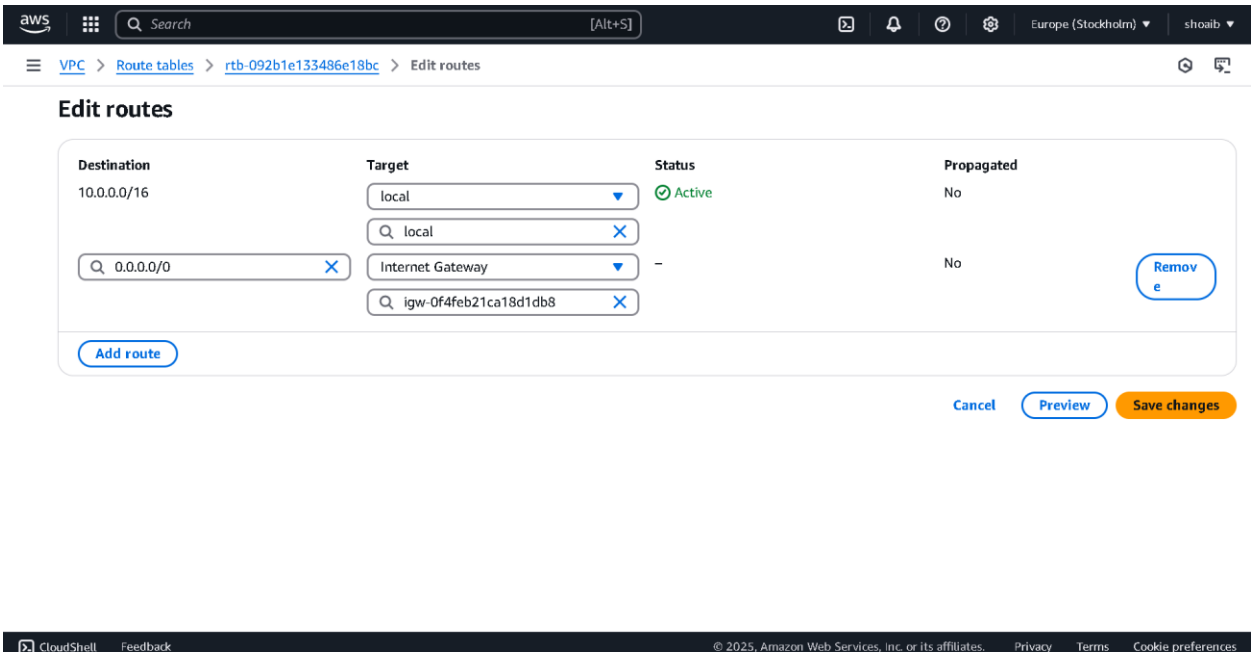


5. Now create route table

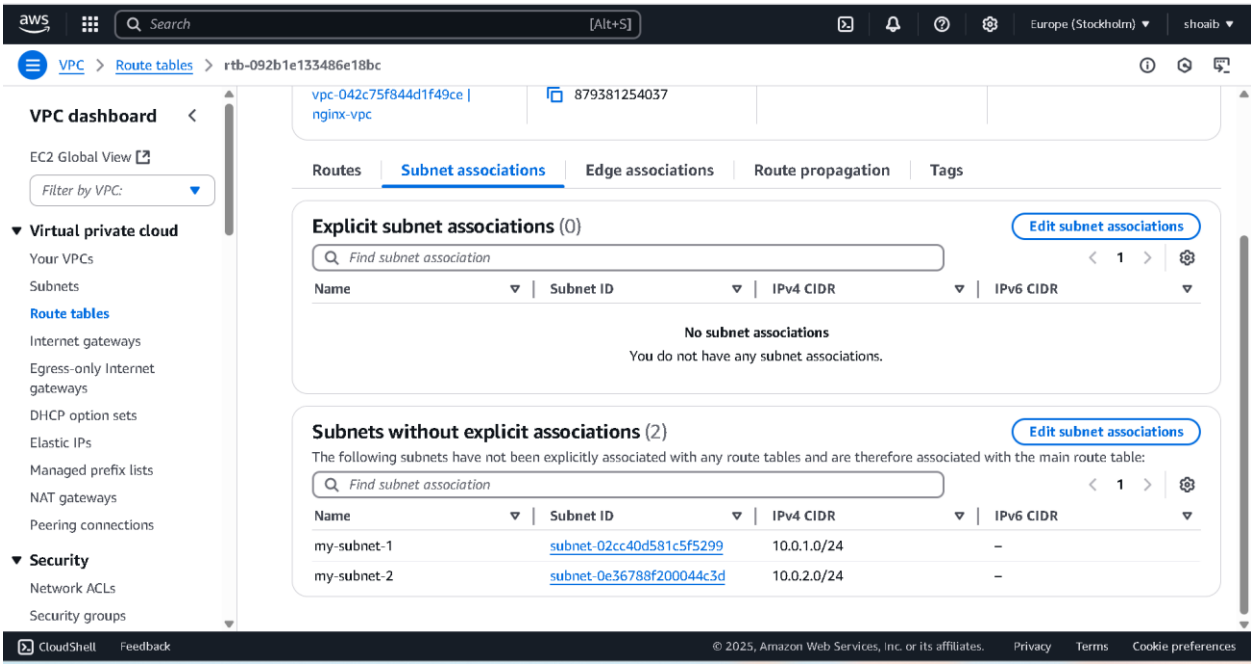
Keep name nginx-RT and select your VPC also

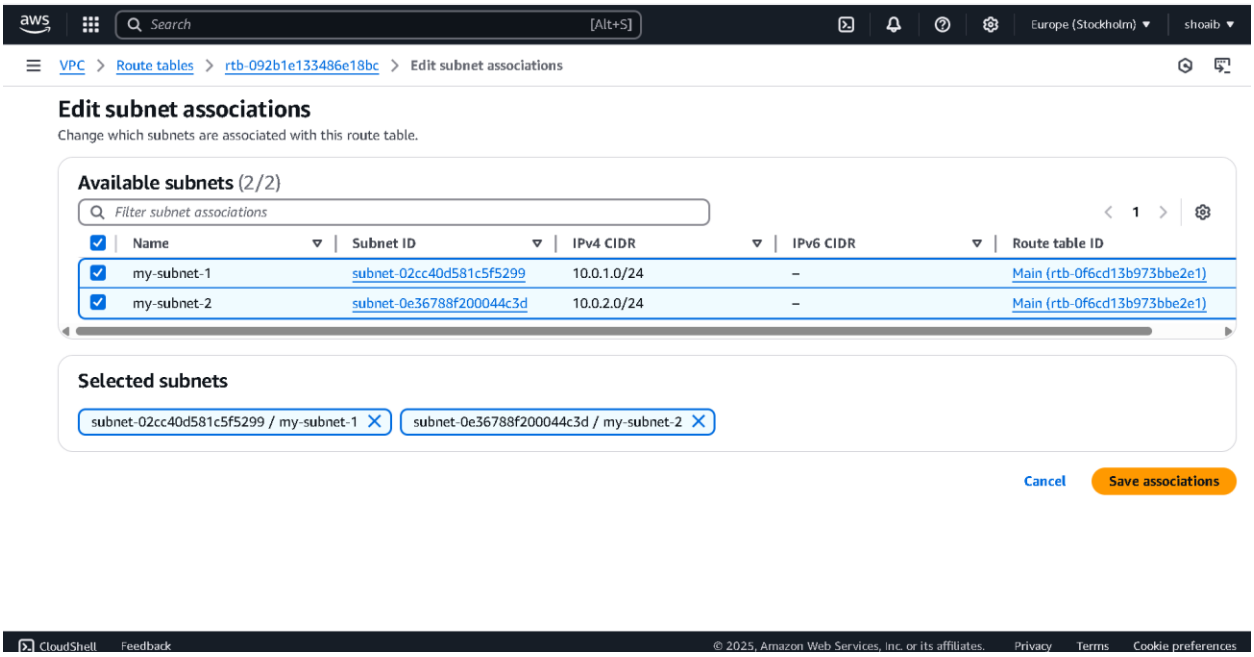


Now click edit routes and add route
And add 0.0.0.0/0 and select your internet gateway



Now go to Subnet Associations Tab and click edit Subnet Associations and select 2 subnets





6. Go to Security Group and keep name nginx-SG and add description like allow Firewall Traffic and select your VPC and add rule in Inbound Rule like

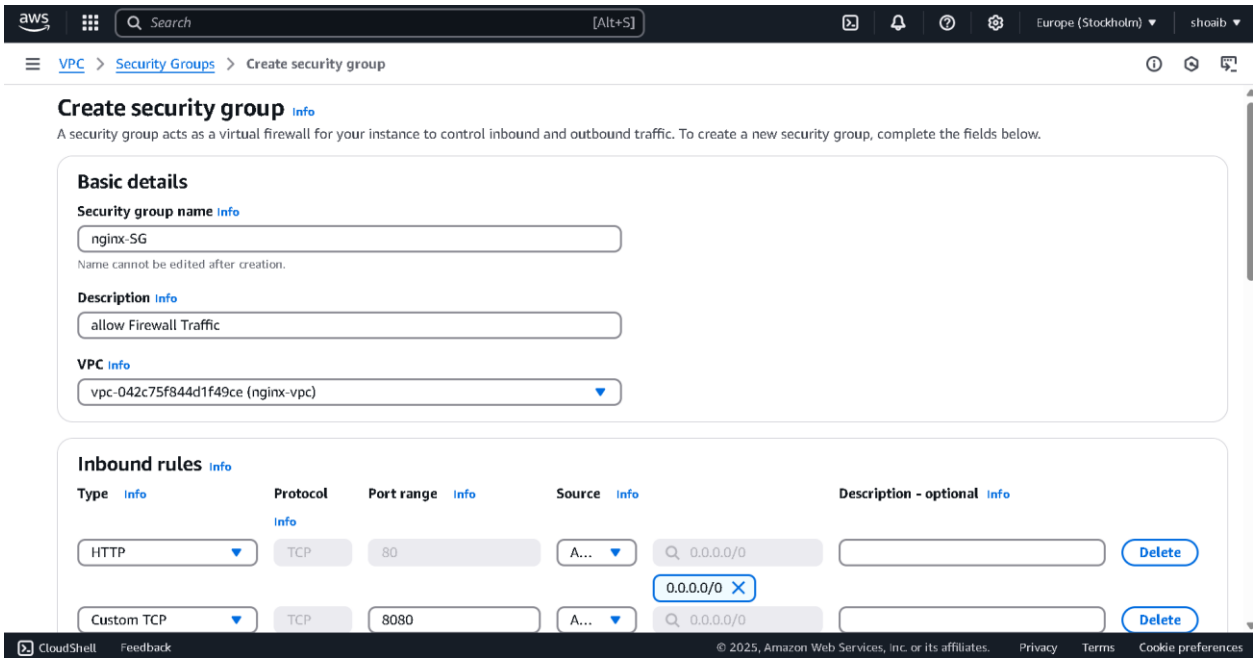
Type=HTTP
PORT=80
SOURCE=anywhere
And add one more rule like

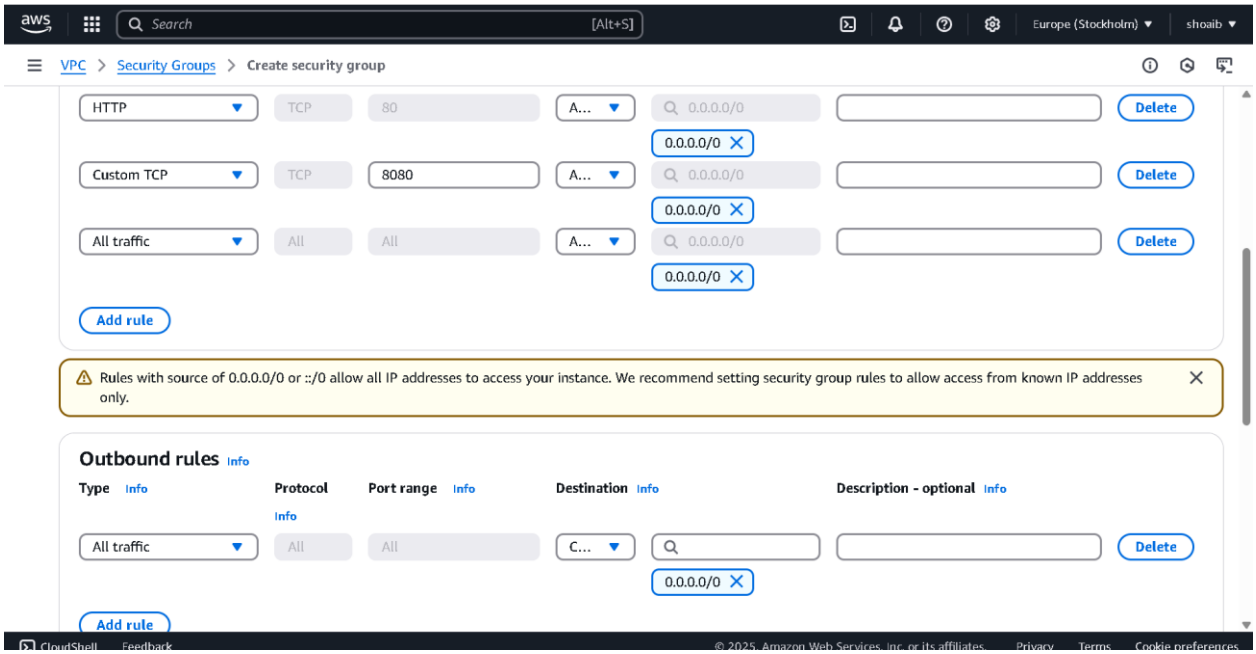
Type=Custom TCP Port
Port range=8080
Source=anywhere

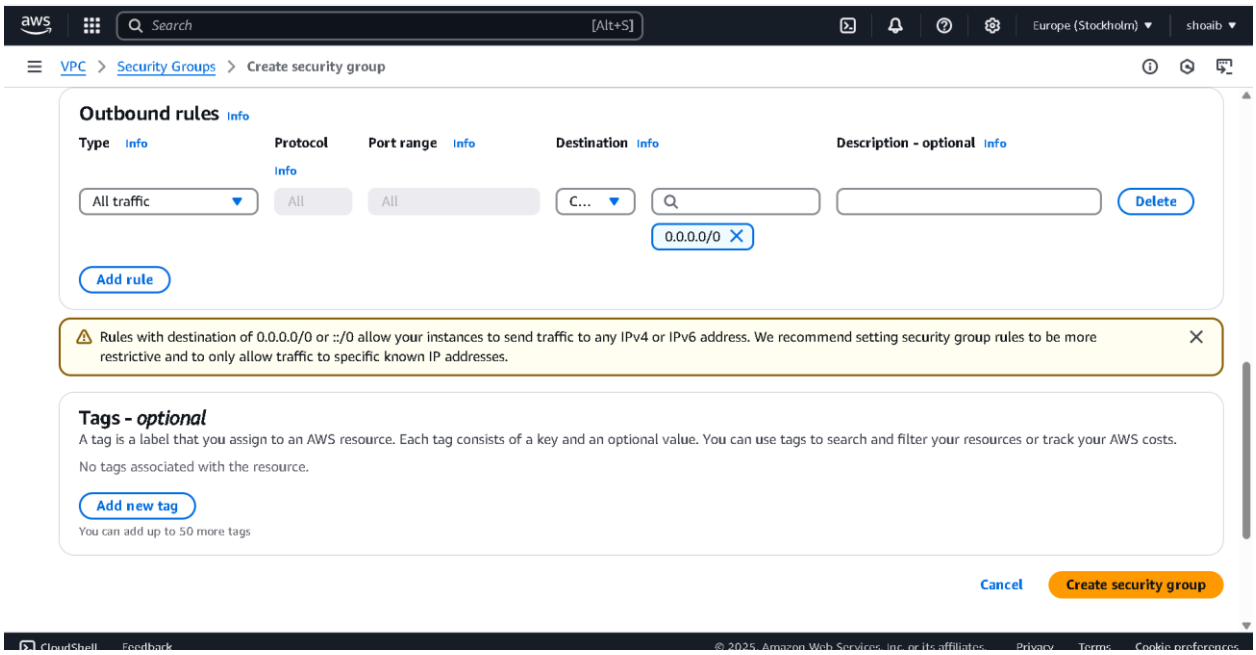
Add one more rule like

Type=All Traffic
Port range=all
Sourse=anywhere

Now click Create Security Group button





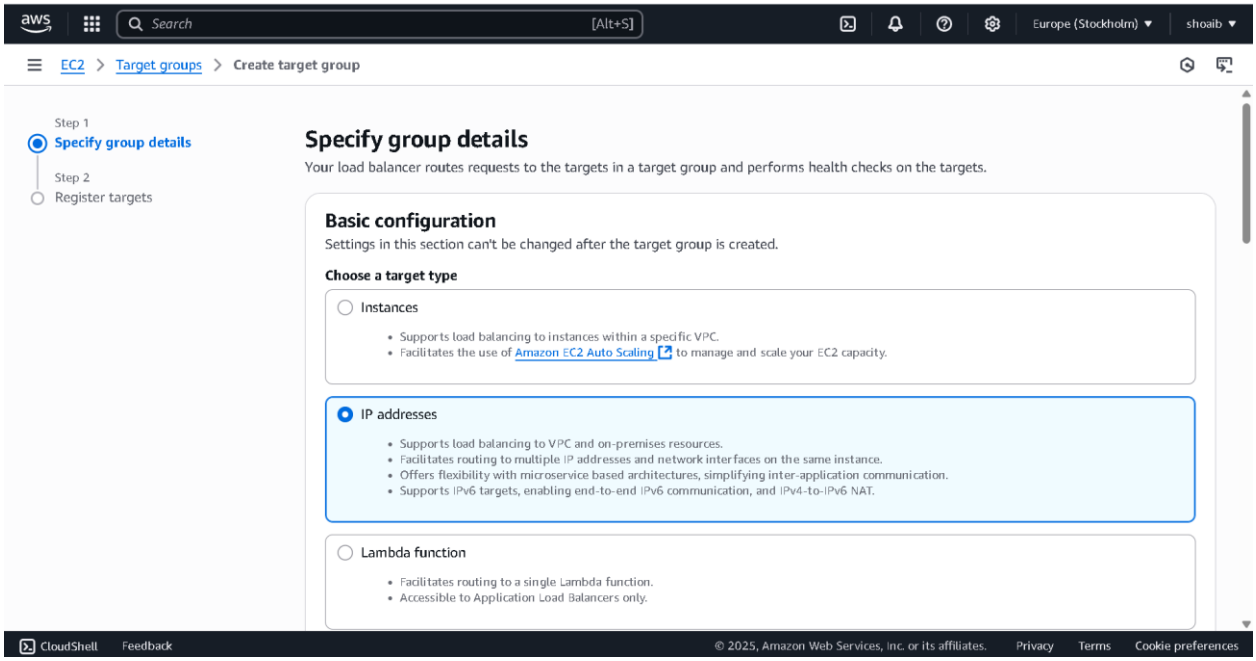


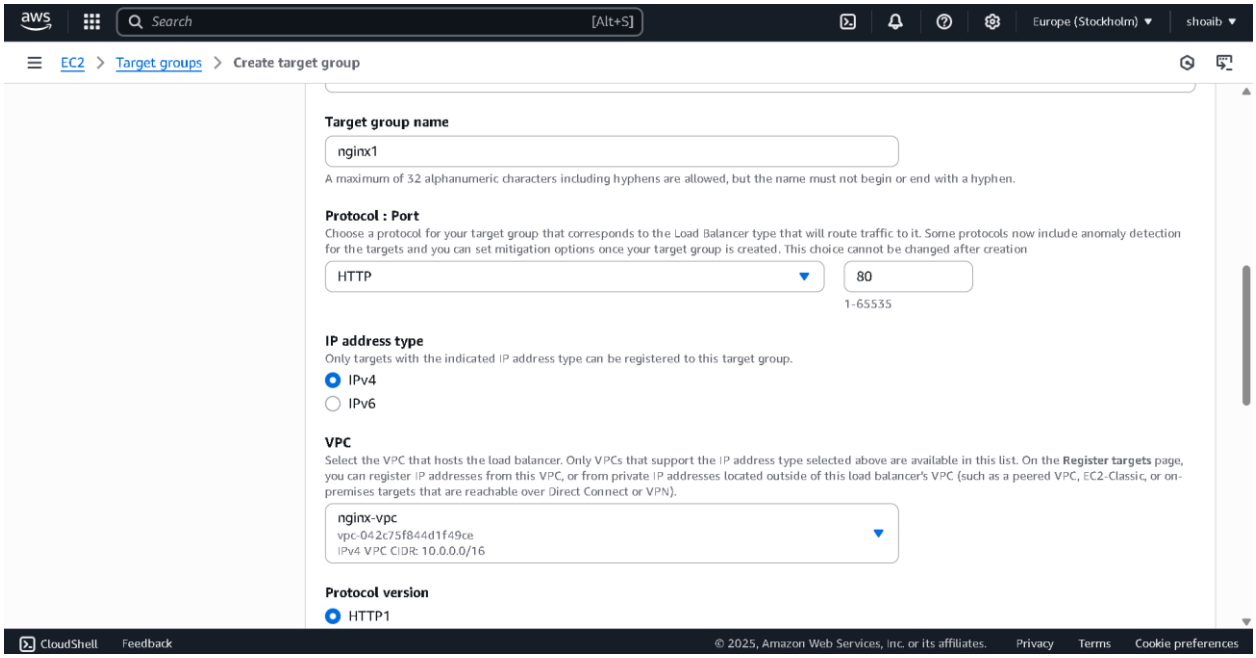
Now go to EC2 and go to Load Balancer
Create Load Balancer and Click Application Load Balancer

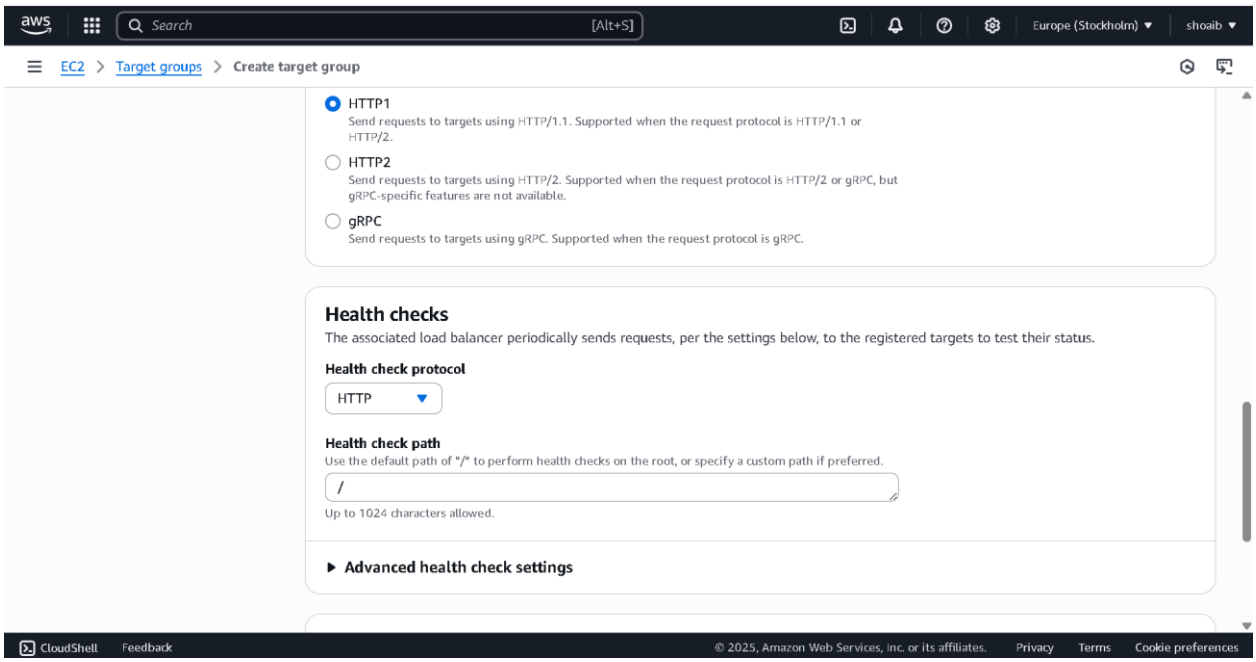
Keep Name Nginx-LB
Select your VPC
Select both **Availability Zones and subnets**
Select your **Security groups**

Now create target group

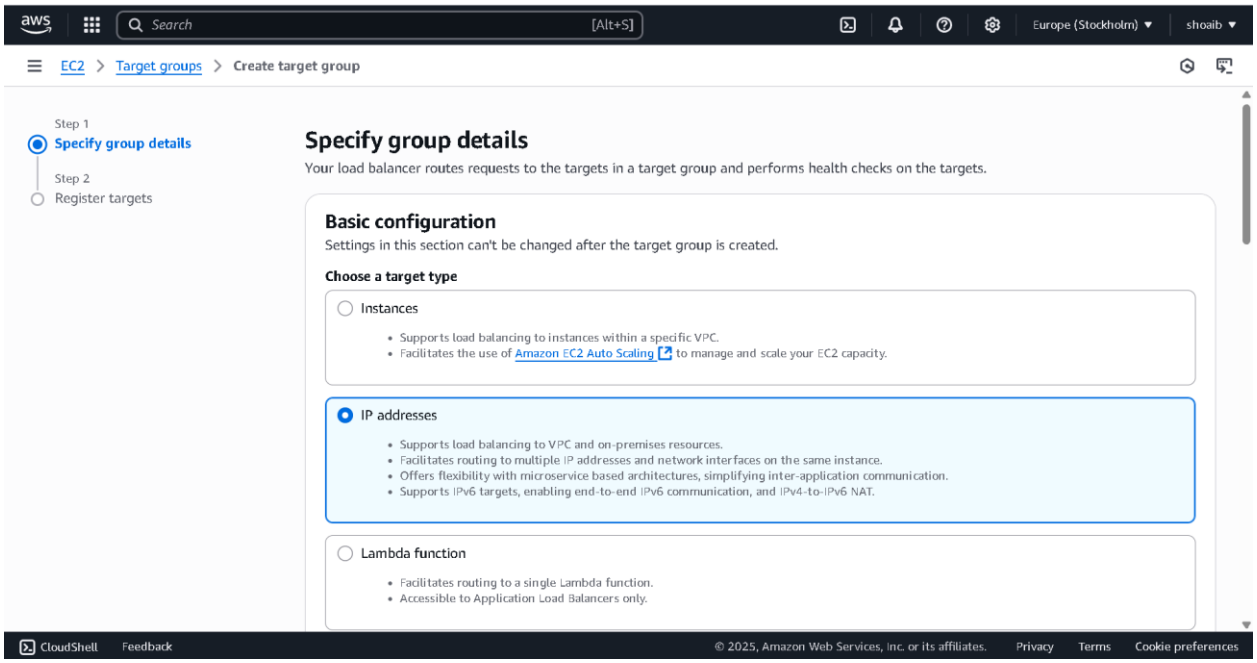
Create Target Group
Select IP addresses
Keep name of target group name as nginx1
Create Target Group by using click button

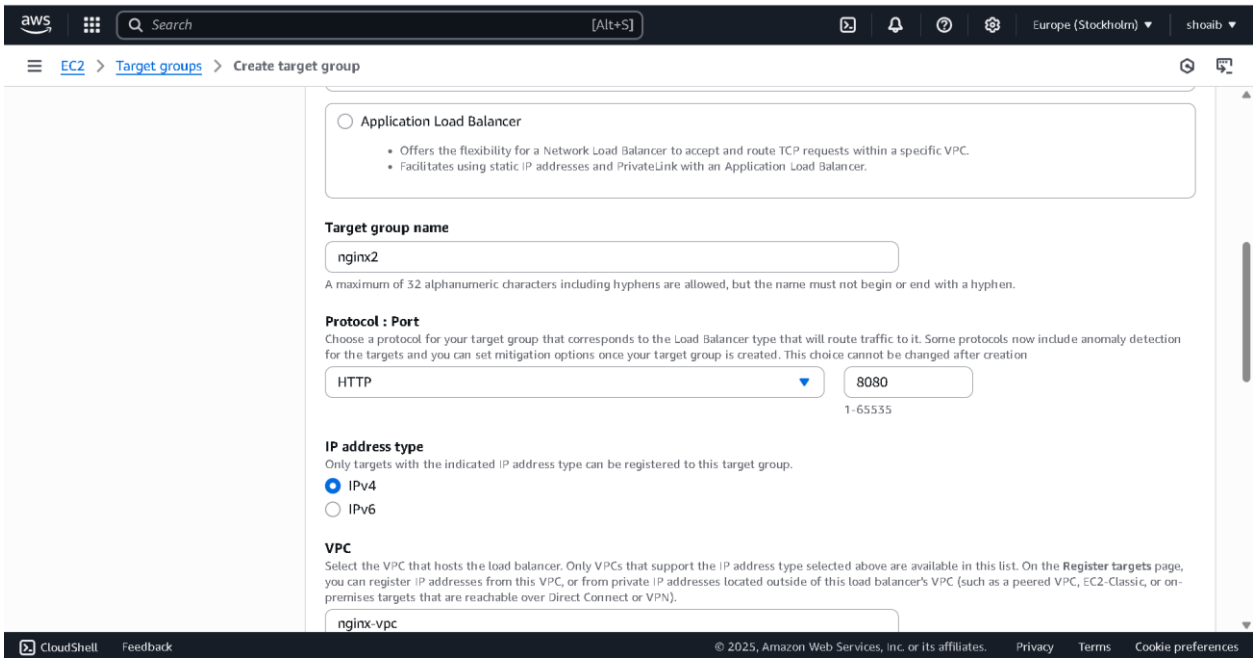


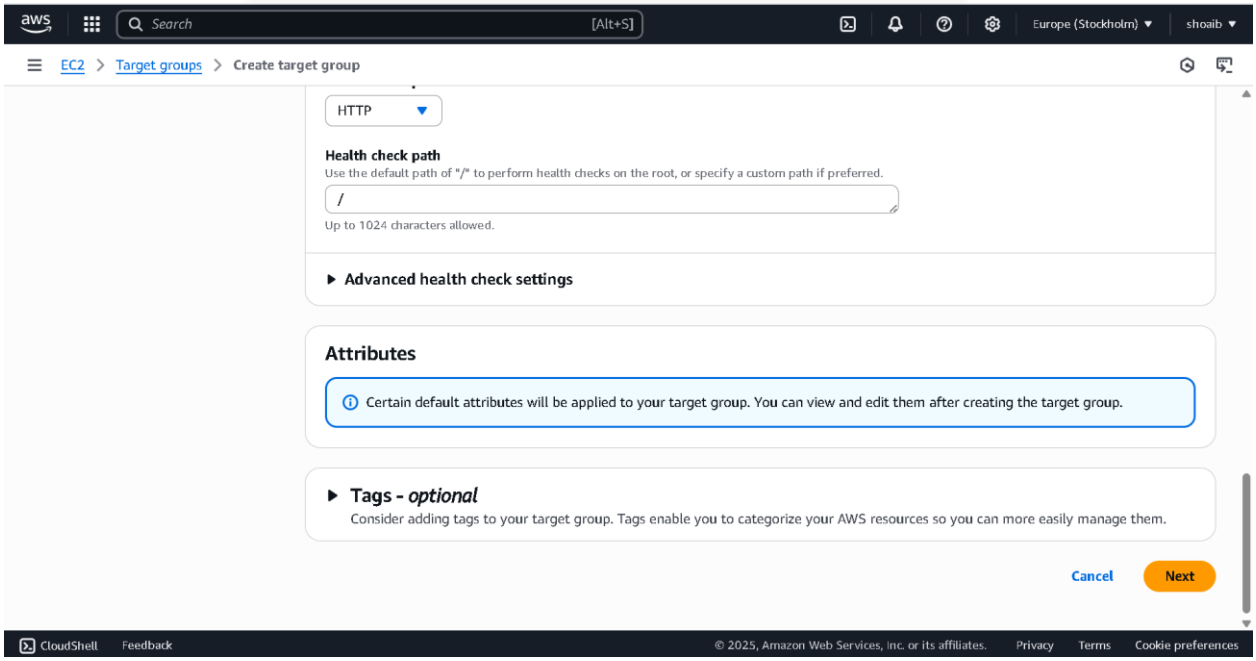


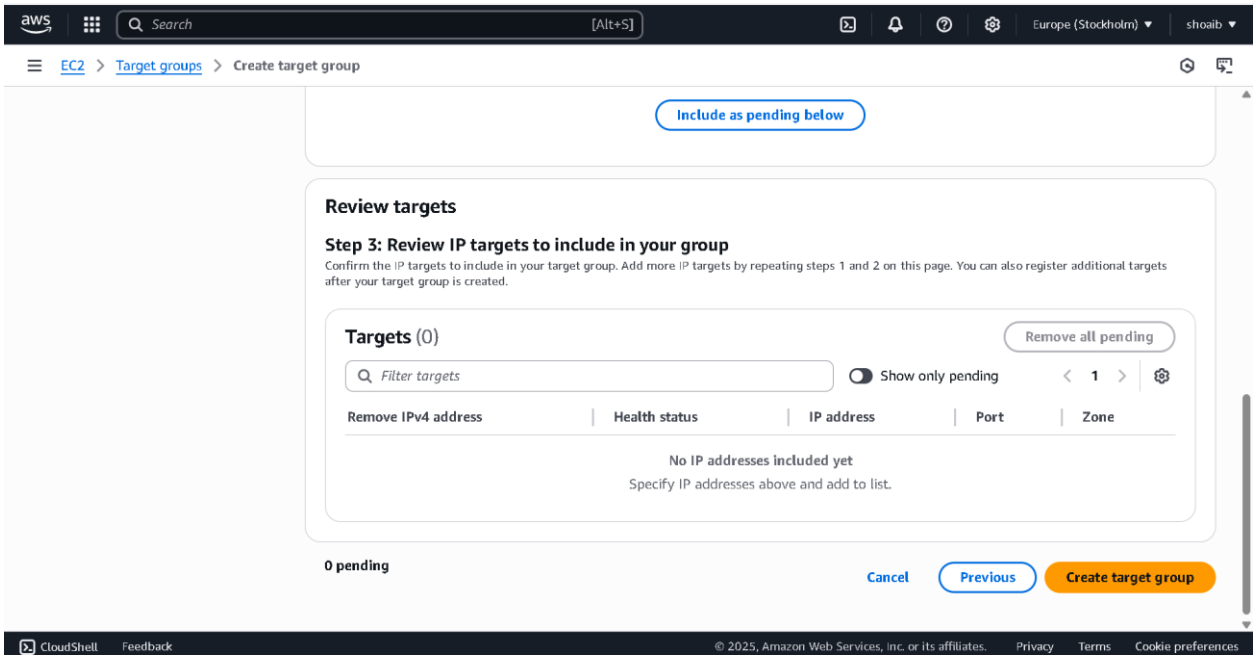


Create second target group
Select IP addresses
Keep name of target group name as nginx2
Keep 8080 port
Select your vpc
Create Target Group by using click button


Search [Alt+S]
Europe (Stockholm)
shoaib
EC2 > Target groups > Create target group
Step 1
Specify group details
Step 2
Register targets
Specify group details
Your load balancer routes requests to the targets in a target group and performs health checks on the targets.
Basic configuration
Settings in this section can't be changed after the target group is created.
Choose a target type
Instances
• Supports load balancing to instances within a specific VPC.
• Facilitates the use of Amazon EC2 Auto Scaling to manage and scale your EC2 capacity.
IP addresses
• Supports load balancing to VPC and on-premises resources.
• Facilitates routing to multiple IP addresses and network interfaces on the same instance.
• Offers flexibility with microservice based architectures, simplifying inter-application communication.
• Supports IPv6 targets, enabling end-to-end IPv6 communication, and IPv4-to-IPv6 NAT.
Lambda function
• Facilitates routing to a single Lambda function.
• Accessible to Application Load Balancers only.
CloudShell
Feedback
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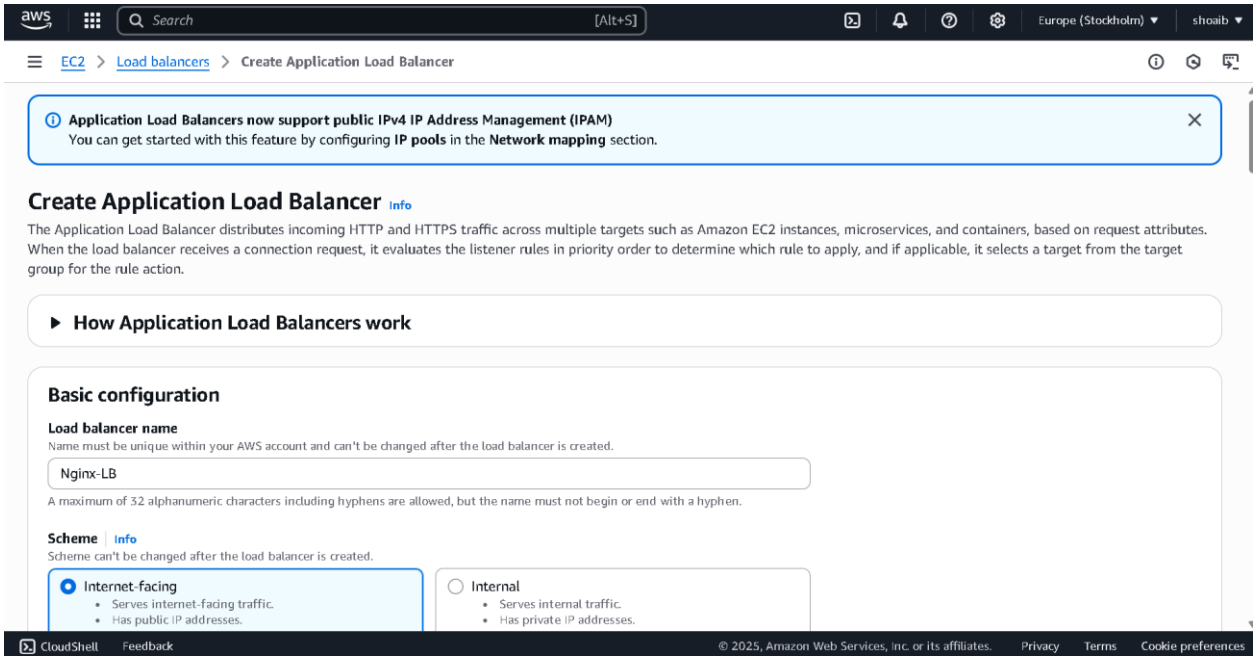

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EC2 > Target groups > Create target group
Application Load Balancer
• Offers the flexibility for a Network Load Balancer to accept and route TCP requests within a specific VPC.
• Facilitates using static IP addresses and PrivateLink with an Application Load Balancer.
Target group name
nginx2
A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.
Protocol : Port
Choose a protocol for your target group that corresponds to the Load Balancer type that will route traffic to it. Some protocols now include anomaly detection for the targets and you can set mitigation options once your target group is created. This choice cannot be changed after creation
HTTP
8080
1-65535
IP address type
Only targets with the indicated IP address type can be registered to this target group.
IPv4
IPv6
VPC
Select the VPC that hosts the load balancer. Only VPCs that support the IP address type selected above are available in this list. On the Register targets page, you can register IP addresses from this VPC, or from private IP addresses located outside of this load balancer's VPC (such as a peered VPC, EC2-Classic, or on-premises targets that are reachable over Direct Connect or VPN).
nginx-vpc
CloudShell
Feedback
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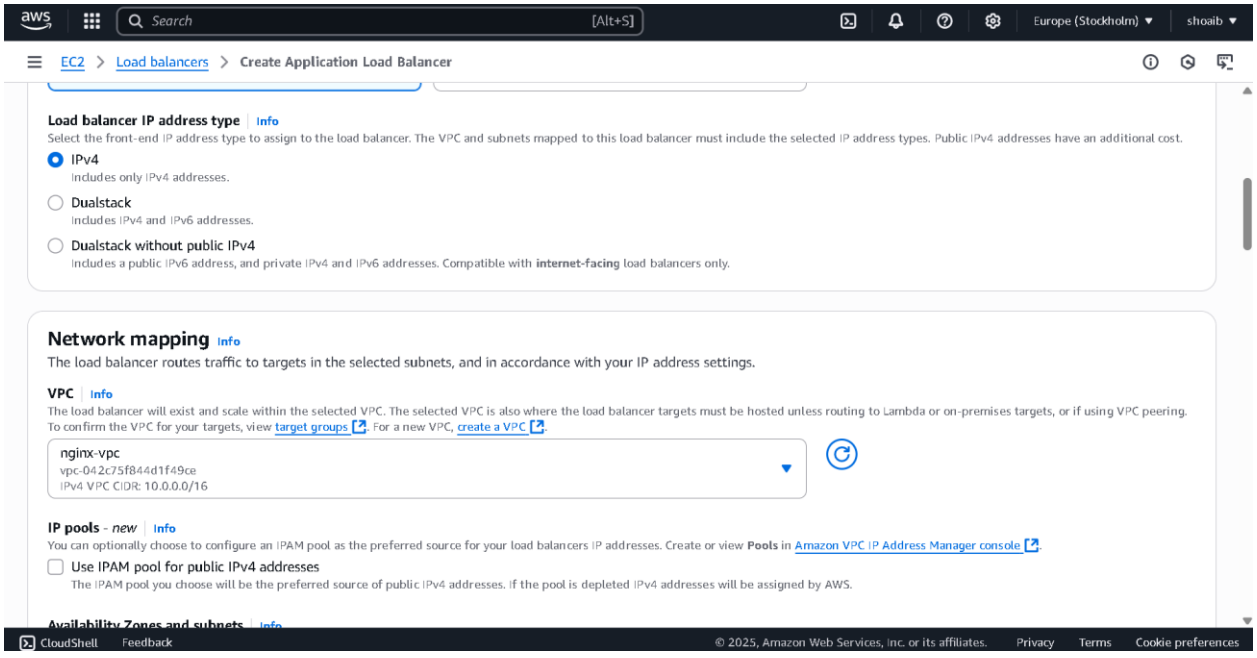


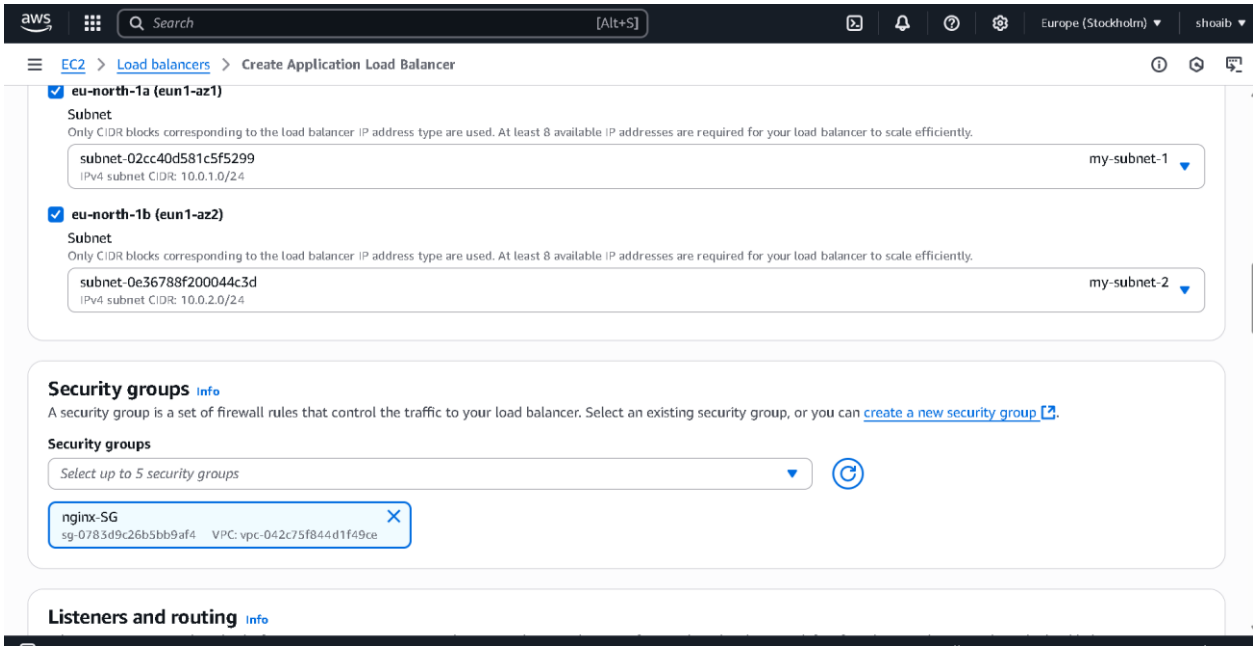


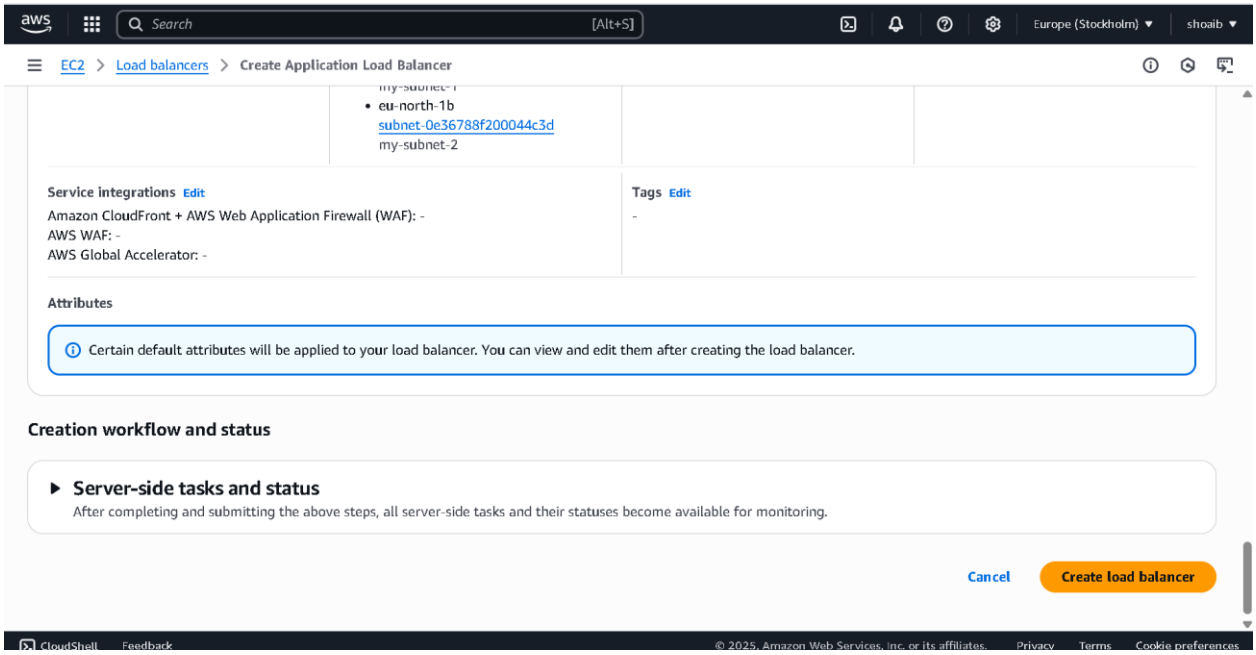
After completed target groups then

Now select nginx1 security group
Create Load Balancer


aws
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EC2 > Load balancers > Create Application Load Balancer
Application Load Balancers now support public IPv4 IP Address Management (IPAM)
You can get started with this feature by configuring IP pools in the Network mapping section.
Create Application Load Balancer Info
The Application Load Balancer distributes incoming HTTP and HTTPS traffic across multiple targets such as Amazon EC2 instances, microservices, and containers, based on request attributes. When the load balancer receives a connection request, it evaluates the listener rules in priority order to determine which rule to apply, and if applicable, it selects a target from the target group for the rule action.
How Application Load Balancers work
Basic configuration
Load balancer name
Name must be unique within your AWS account and can't be changed after the load balancer is created.
Nginx-LB
A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.
Scheme Info
Scheme can't be changed after the load balancer is created.
Internet-facing
Serves internet-facing traffic.
Has public IP addresses.
Internal
Serves internal traffic.
Has private IP addresses.
CloudShell
Feedback
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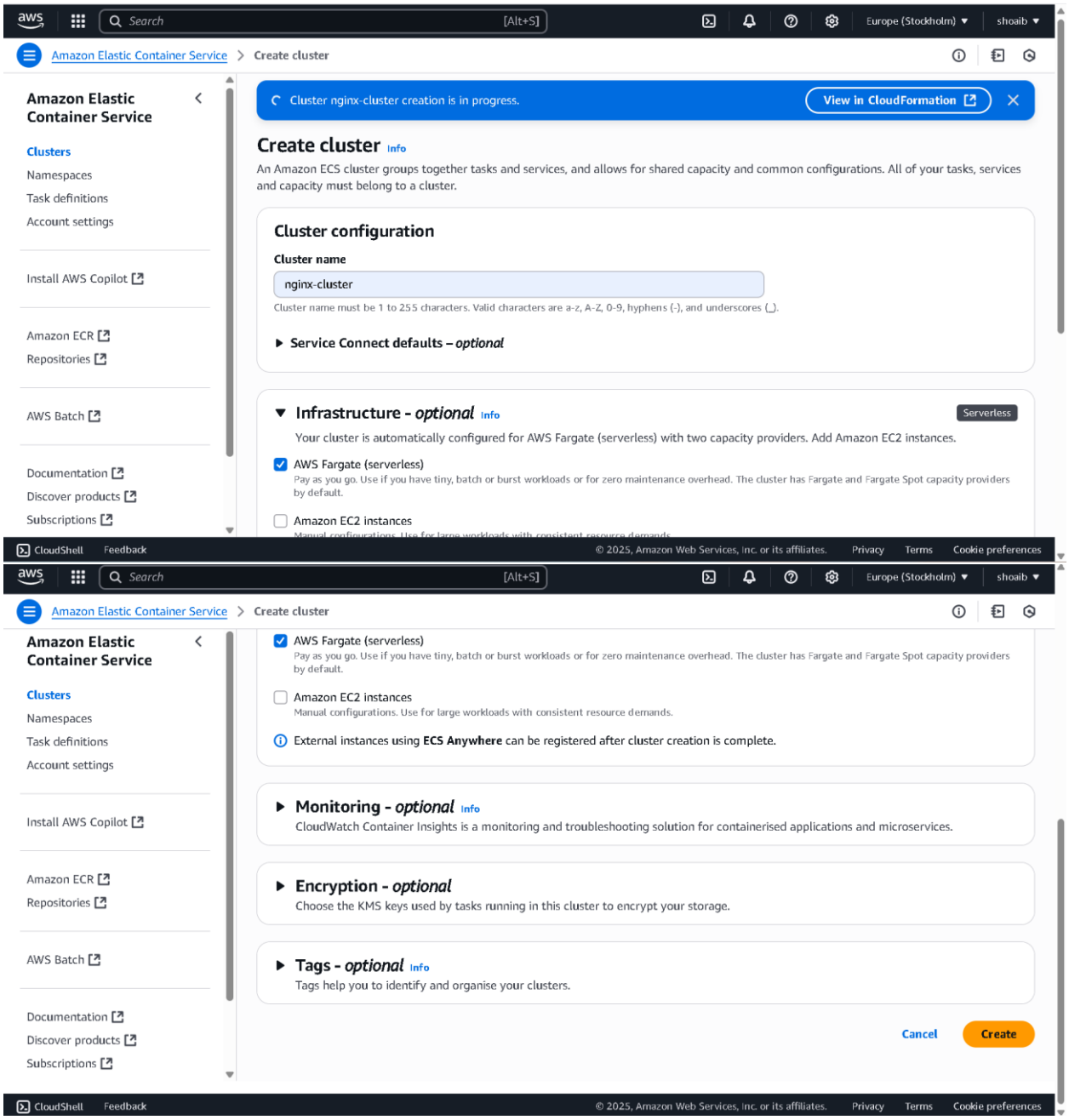

aws
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EC2 > Load balancers > Create Application Load Balancer
Load balancer IP address type Info
Select the front-end IP address type to assign to the load balancer. The VPC and subnets mapped to this load balancer must include the selected IP address types. Public IPv4 addresses have an additional cost.
IPv4
Includes only IPv4 addresses.
Dualstack
Includes IPv4 and IPv6 addresses.
Dualstack without public IPv4
Includes a public IPv6 address, and private IPv4 and IPv6 addresses. Compatible with internet-facing load balancers only.
Network mapping Info
The load balancer routes traffic to targets in the selected subnets, and in accordance with your IP address settings.
VPC Info
The load balancer will exist and scale within the selected VPC. The selected VPC is also where the load balancer targets must be hosted unless routing to Lambda or on-premises targets, or if using VPC peering. To confirm the VPC for your targets, view target groups. For a new VPC, create a VPC.
nginx-vpc
vpc-042c75f844d1f49ce
IPv4 VPC CIDR: 10.0.0.0/16
IP pools - new Info
You can optionally choose to configure an IPAM pool as the preferred source for your load balancers IP addresses. Create or view Pools in Amazon VPC IP Address Manager console.
Use IPAM pool for public IPv4 addresses
The IPAM pool you choose will be the preferred source of public IPv4 addresses. If the pool is depleted IPv4 addresses will be assigned by AWS.
Availability Zones and subnets Info
CloudShell
Feedback
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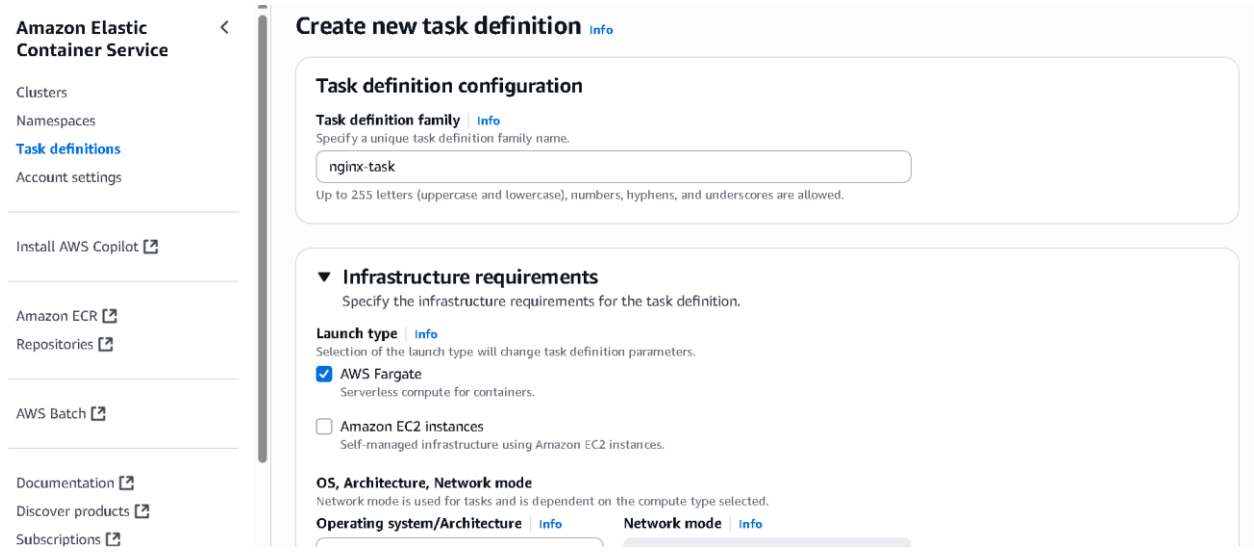
Now go to ECS

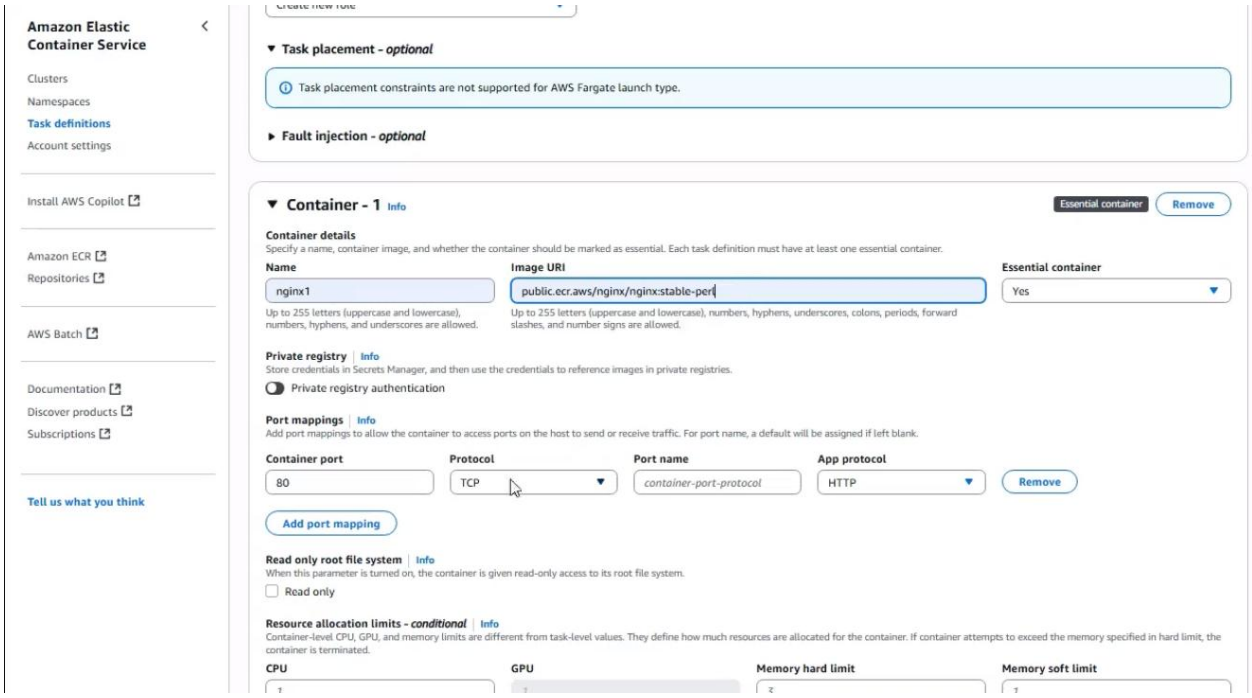
Select Cluster and keep name as nginx-cluster
Select AWS Fragat

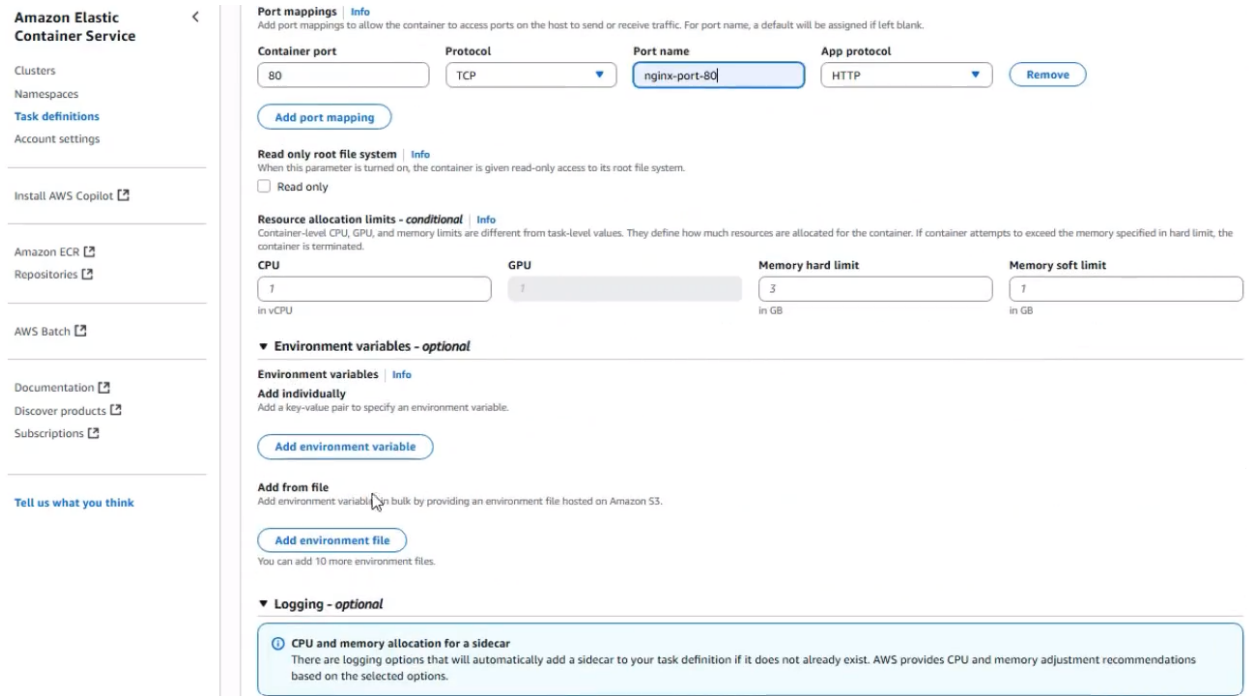


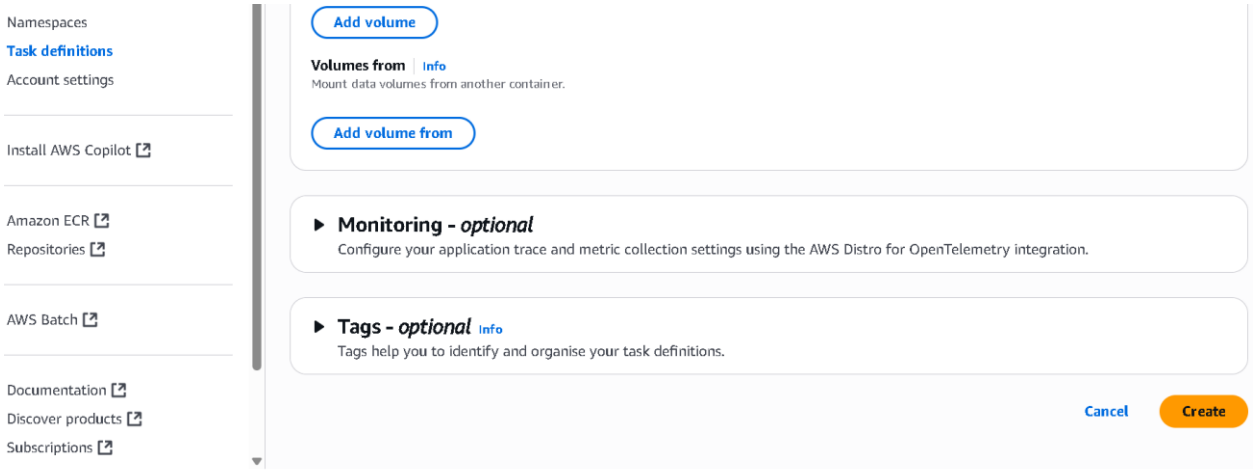
Now go to Task Defination and create Task Definition

Write name like nginx-task
Give name of container and write image url
Write port name also
Then click create button


Amazon Elastic Container Service
Clusters
Namespaces
Task definitions
Account settings
Install AWS Copilot
Amazon ECR
Repositories
AWS Batch
Documentation
Discover products
Subscriptions
Create new task definition Info
Task definition configuration
Task definition family | Info
Specify a unique task definition family name.
nginx-task
Up to 255 letters (uppercase and lowercase), numbers, hyphens, and underscores are allowed.
Infrastructure requirements
Specify the infrastructure requirements for the task definition.
Launch type | Info
Selection of the launch type will change task definition parameters.
AWS Fargate
Serverless compute for containers.
Amazon EC2 instances
Self-managed infrastructure using Amazon EC2 instances.
OS, Architecture, Network mode
Network mode is used for tasks and is dependent on the compute type selected.
Operating system/Architecture | Info
Network mode | Info

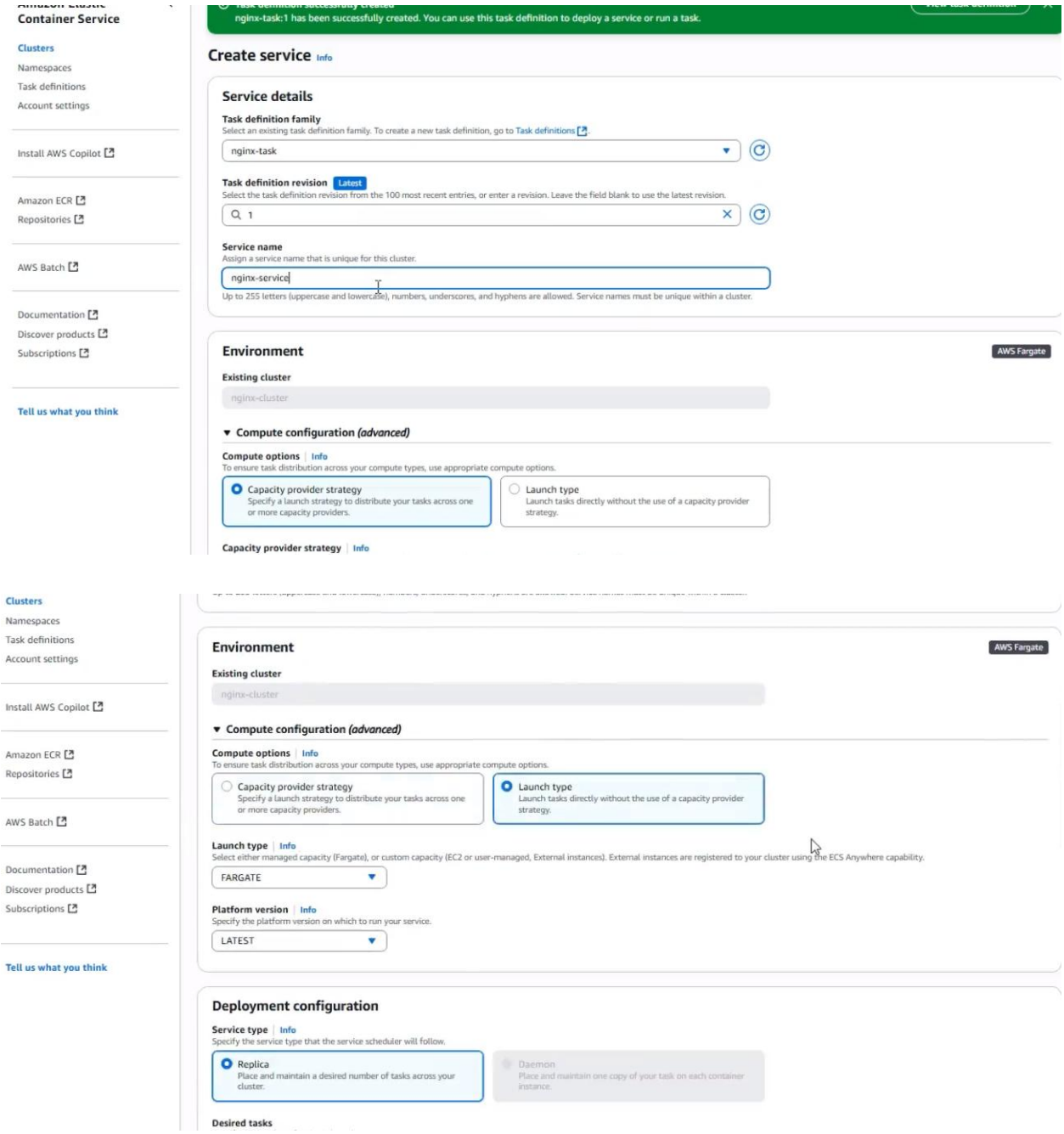

Amazon Elastic Container Service
Clusters
Namespaces
Task definitions
Account settings
Install AWS Copilot
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Repositories
AWS Batch
Documentation
Discover products
Subscriptions
Tell us what you think
Task placement - optional
Task placement constraints are not supported for AWS Fargate launch type.
Fault injection - optional
Container - 1 Info
Essential container
Remove
Container details
Specify a name, container image, and whether the container should be marked as essential. Each task definition must have at least one essential container.
Name
nginx1
Up to 255 letters (uppercase and lowercase), numbers, hyphens, and underscores are allowed.
Image URI
public.ecr.aws/nginx/nginx:stable-perl
Up to 255 letters (uppercase and lowercase), numbers, hyphens, underscores, colons, periods, forward slashes, and number signs are allowed.
Essential container
Yes
Private registry | Info
Store credentials in Secrets Manager, and then use the credentials to reference images in private registries.
Private registry authentication
Port mappings | Info
Add port mappings to allow the container to access ports on the host to send or receive traffic. For port name, a default will be assigned if left blank.
Container port
80
Protocol
TCP
Port name
container-port-protocol
App protocol
HTTP
Remove
Add port mapping
Read only root file system | Info
When this parameter is turned on, the container is given read-only access to its root file system.
Read only
Resource allocation limits - conditional | Info
Container-level CPU, GPU, and memory limits are different from task-level values. They define how much resources are allocated for the container. If container attempts to exceed the memory specified in hard limit, the container is terminated.
CPU
GPU
Memory hard limit
Memory soft limit

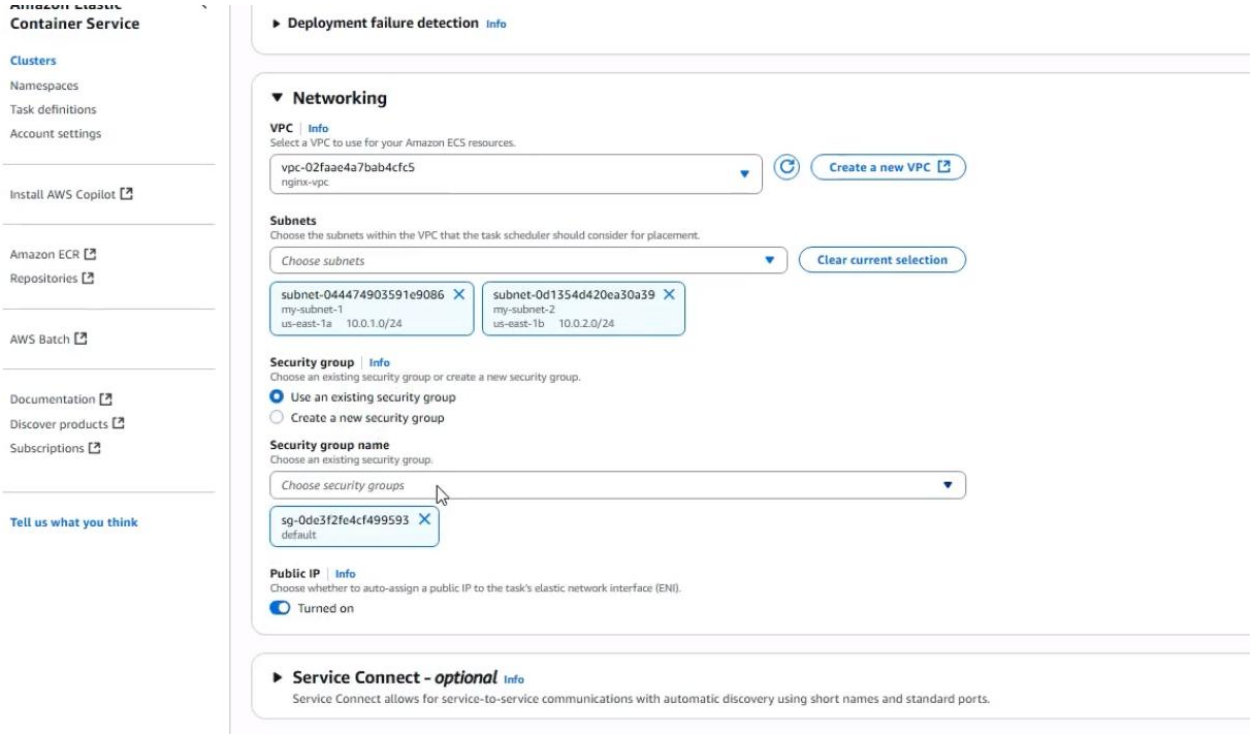


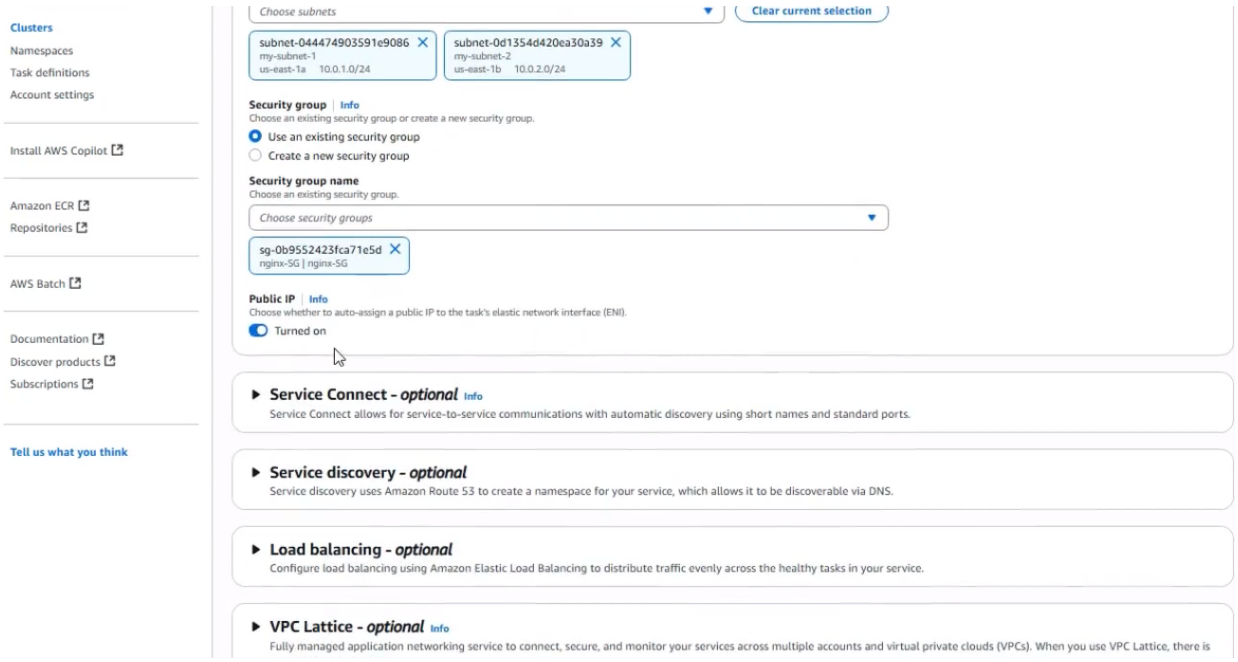


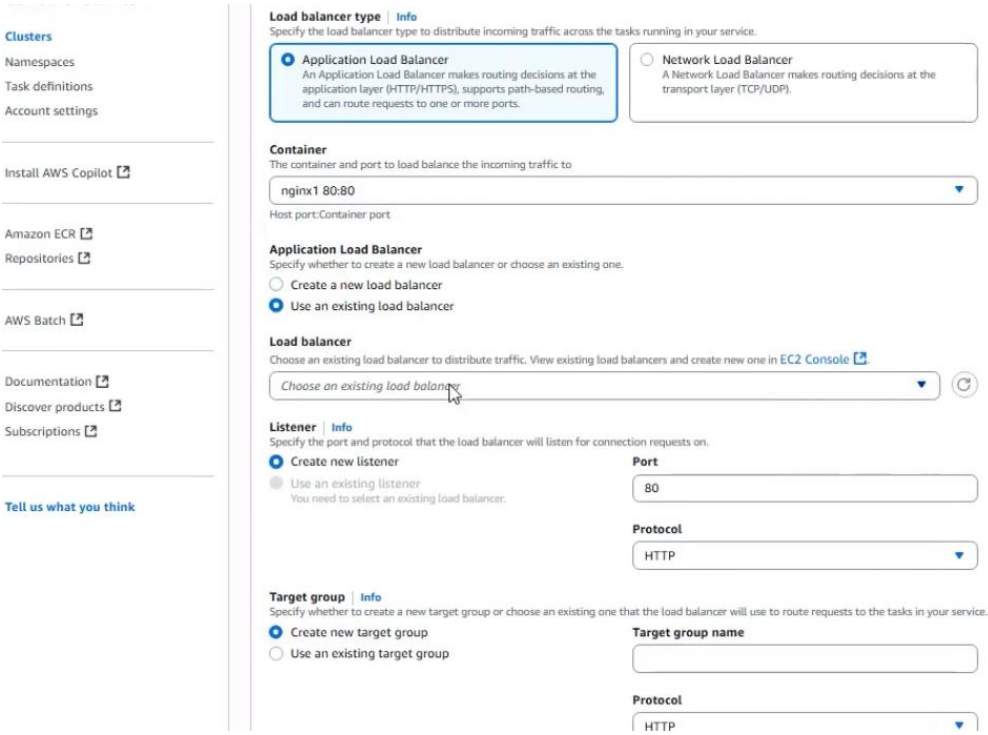
Now go to cluster and create service

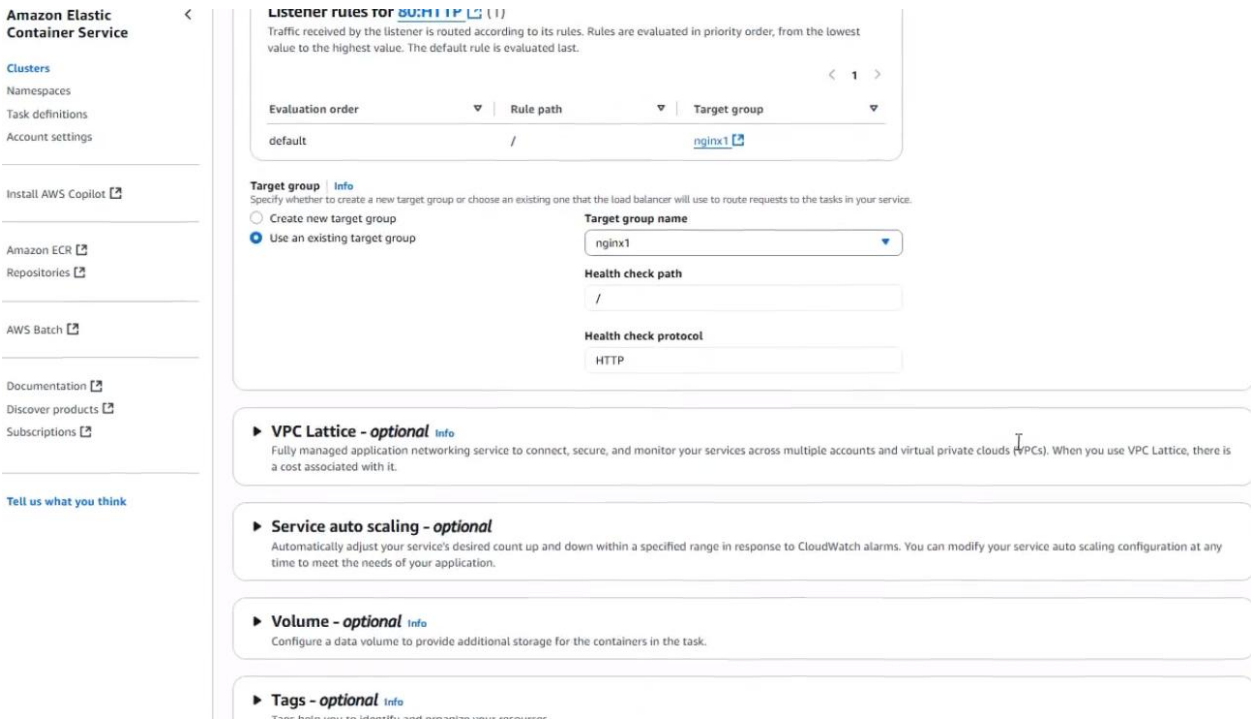
Write Task Definition Family Name like nginx-task
Keep service name like nginx-service
Select Launch Type under Compute Options
Select your VPC under Networking
Select your own security group
Keep your own load balancer
Now create service by using clicking button

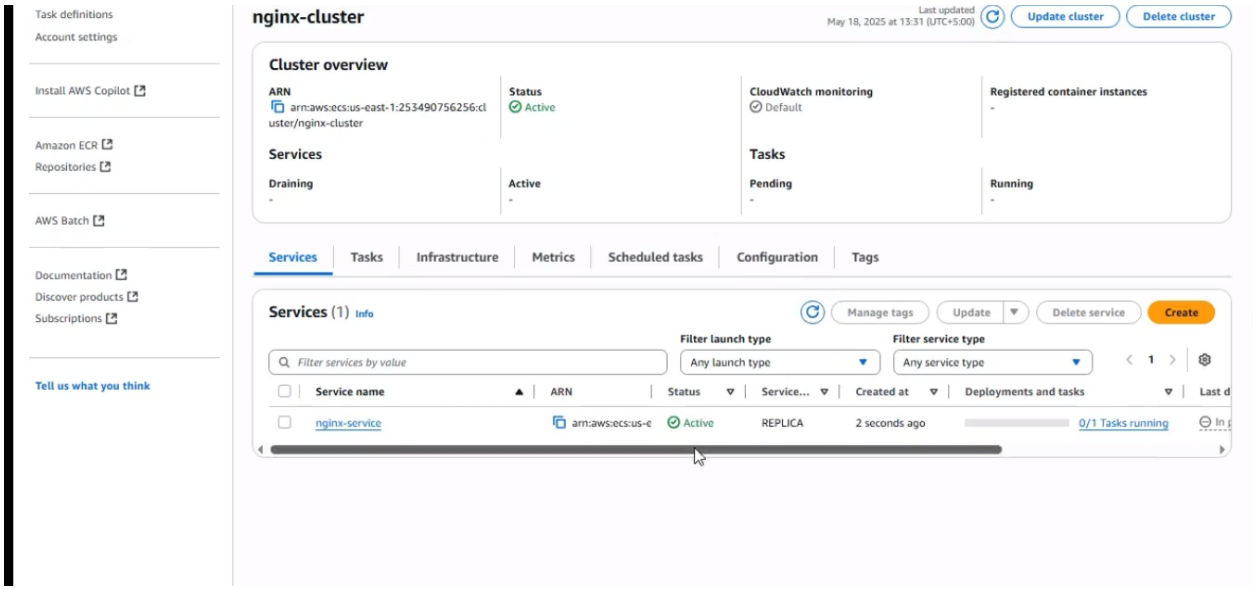

Container Service
Clusters
Namespaces
Task definitions
Account settings
Install AWS Copilot
Amazon ECR
Repositories
AWS Batch
Documentation
Discover products
Subscriptions
Tell us what you think
nginx-task:1 has been successfully created. You can use this task definition to deploy a service or run a task.
Create service Info
Service details
Task definition family
Select an existing task definition family. To create a new task definition, go to Task definitions.
nginx-task
Task definition revision Latest
Select the task definition revision from the 100 most recent entries, or enter a revision. Leave the field blank to use the latest revision.
1
Service name
Assign a service name that is unique for this cluster.
nginx-service
Up to 255 letters (uppercase and lowercase), numbers, underscores, and hyphens are allowed. Service names must be unique within a cluster.
Environment
AWS Fargate
Existing cluster
nginx-cluster
Compute configuration (advanced)
Compute options | Info
To ensure task distribution across your compute types, use appropriate compute options.
Capacity provider strategy
Specify a launch strategy to distribute your tasks across one or more capacity providers.
Launch type
Launch tasks directly without the use of a capacity provider strategy.
Capacity provider strategy | Info
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Launch tasks directly without the use of a capacity provider strategy.
Launch type | Info
Select either managed capacity (Fargate), or custom capacity (EC2 or user-managed, External instances). External instances are registered to your cluster using the ECS Anywhere capability.
FARGATE
Platform version | Info
Specify the platform version on which to run your service.
LATEST
Deployment configuration
Service type | Info
Specify the service type that the service scheduler will follow.
Replica
Place and maintain a desired number of tasks across your cluster.
Daemon
Place and maintain one copy of your task on each container instance.
Desired tasks


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Deployment failure detection Info
Networking
VPC | Info
Select a VPC to use for your Amazon ECS resources.
vpc-02faae4a7bab4cfc5
nginx-vpc
Create a new VPC
Subnets
Choose the subnets within the VPC that the task scheduler should consider for placement.
Choose subnets
Clear current selection
subnet-044474903591e9086
my-subnet-1
us-east-1a 10.0.1.0/24
subnet-0d1354d420ea30a39
my-subnet-2
us-east-1b 10.0.2.0/24
Security group | Info
Choose an existing security group or create a new security group.
Use an existing security group
Create a new security group
Security group name
Choose an existing security group.
Choose security groups
sg-0de3f2fe4cf499593
default
Public IP | Info
Choose whether to auto-assign a public IP to the task's elastic network interface (ENI).
Turned on
Service Connect - optional Info
Service Connect allows for service-to-service communications with automatic discovery using short names and standard ports.


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Choose subnets
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my-subnet-2
us-east-1b 10.0.2.0/24
Security group | Info
Choose an existing security group or create a new security group.
Use an existing security group
Create a new security group
Security group name
Choose an existing security group.
Choose security groups
sg-0b9552423fca71e5d
nginx-SG | nginx-SG
Public IP | Info
Choose whether to auto-assign a public IP to the task's elastic network interface (ENI).
Turned on
Service Connect - optional Info
Service Connect allows for service-to-service communications with automatic discovery using short names and standard ports.
Service discovery - optional
Service discovery uses Amazon Route 53 to create a namespace for your service, which allows it to be discoverable via DNS.
Load balancing - optional
Configure load balancing using Amazon Elastic Load Balancing to distribute traffic evenly across the healthy tasks in your service.
VPC Lattice - optional Info
Fully managed application networking service to connect, secure, and monitor your services across multiple accounts and virtual private clouds (VPCs). When you use VPC Lattice, there is


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Load balancer type | Info
Specify the load balancer type to distribute incoming traffic across the tasks running in your service.
Application Load Balancer
An Application Load Balancer makes routing decisions at the application layer (HTTP/HTTPS), supports path-based routing, and can route requests to one or more ports.
Network Load Balancer
A Network Load Balancer makes routing decisions at the transport layer (TCP/UDP).
Container
The container and port to load balance the incoming traffic to
nginx1 80:80
Host port:Container port
Application Load Balancer
Specify whether to create a new load balancer or choose an existing one.
Create a new load balancer
Use an existing load balancer
Load balancer
Choose an existing load balancer to distribute traffic. View existing load balancers and create new one in EC2 Console.
Choose an existing load balancer
Listener | Info
Specify the port and protocol that the load balancer will listen for connection requests on.
Create new listener
Use an existing listener
You need to select an existing load balancer.
Port
80
Protocol
HTTP
Target group | Info
Specify whether to create a new target group or choose an existing one that the load balancer will use to route requests to the tasks in your service.
Create new target group
Use an existing target group
Target group name
Protocol
HTTP


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Listener rules for 80:HTTP
Traffic received by the listener is routed according to its rules. Rules are evaluated in priority order, from the lowest value to the highest value. The default rule is evaluated last.
Evaluation order
Rule path
Target group
default
/
nginx1
Target group | Info
Specify whether to create a new target group or choose an existing one that the load balancer will use to route requests to the tasks in your service.
Create new target group
Use an existing target group
Target group name
nginx1
Health check path
/
Health check protocol
HTTP
VPC Lattice - optional Info
Fully managed application networking service to connect, secure, and monitor your services across multiple accounts and virtual private clouds (VPCs). When you use VPC Lattice, there is a cost associated with it.
Service auto scaling - optional
Automatically adjust your service's desired count up and down within a specified range in response to CloudWatch alarms. You can modify your service auto scaling configuration at any time to meet the needs of your application.
Volume - optional Info
Configure a data volume to provide additional storage for the containers in the task.
Tags - optional Info



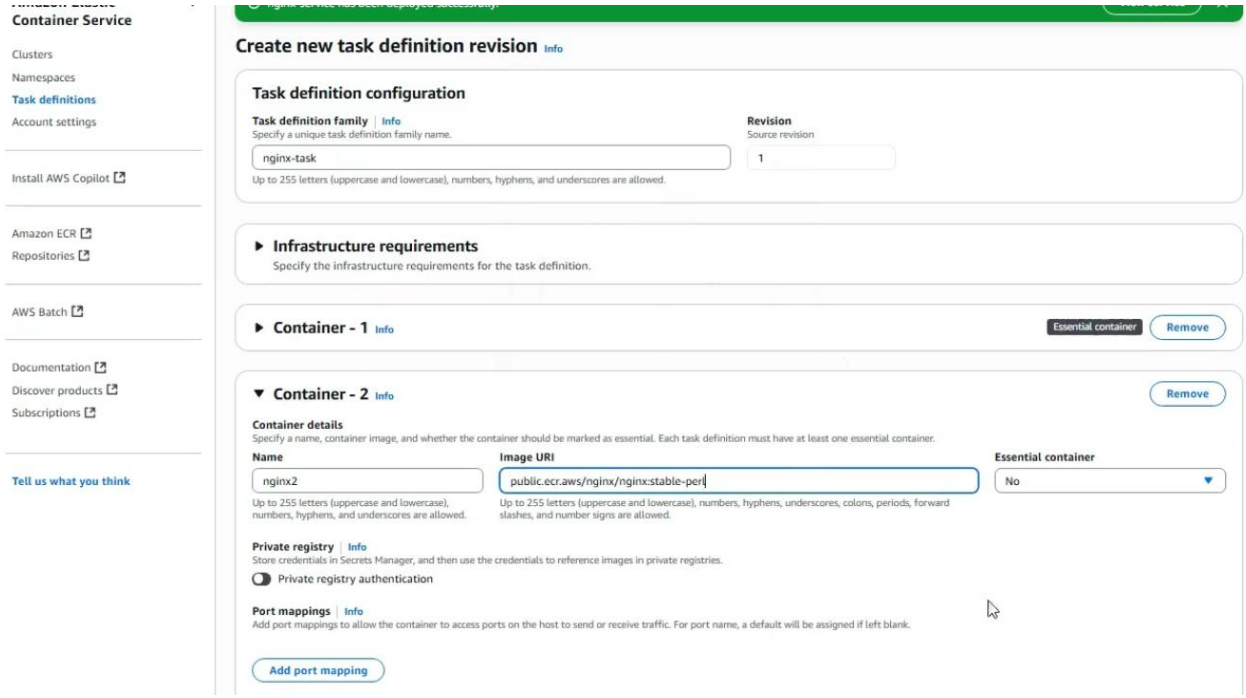
Now go to load balancer and copy name and paste into url

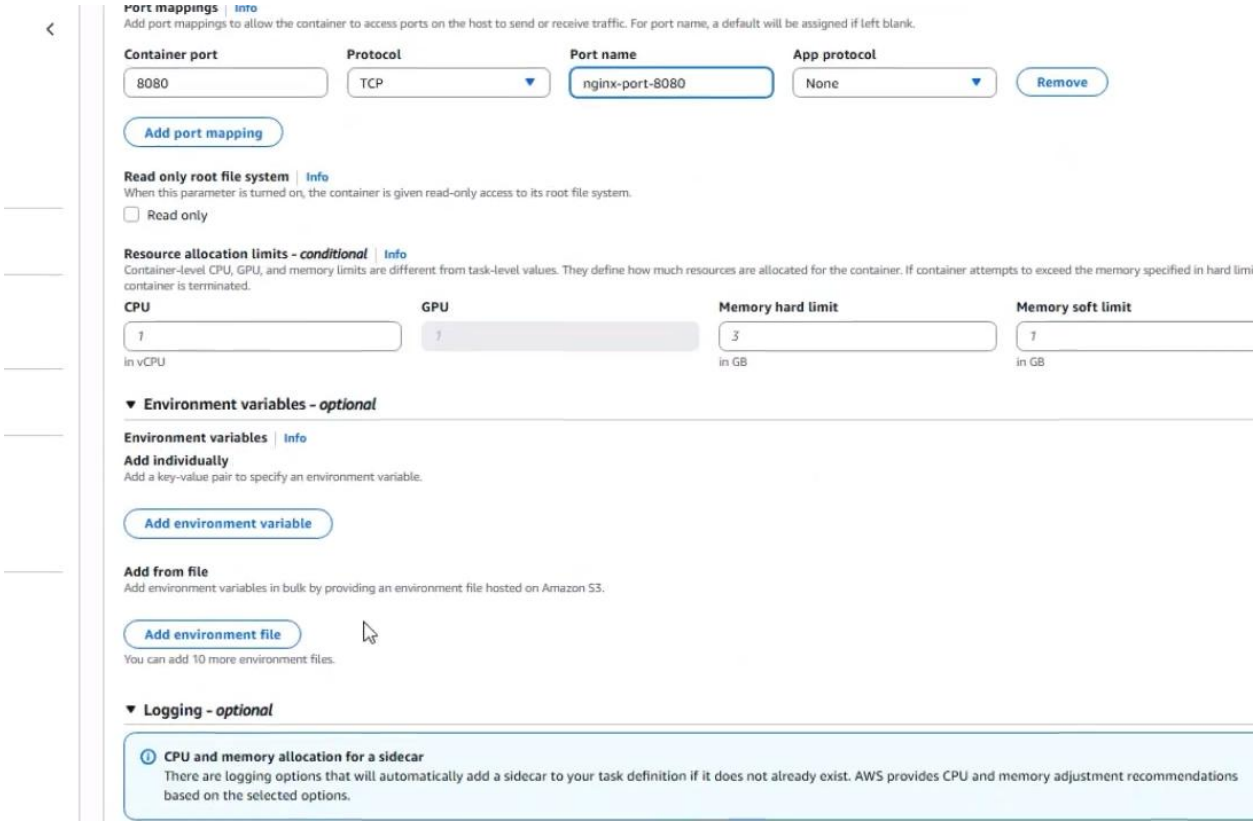
First Output:

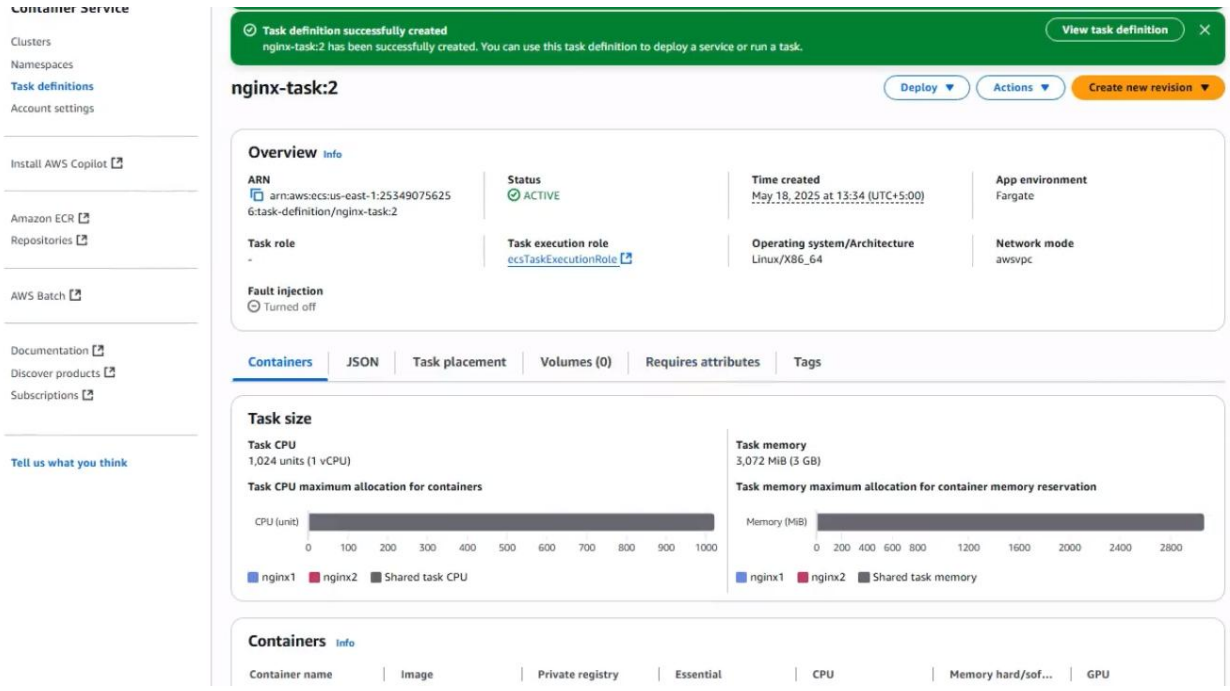


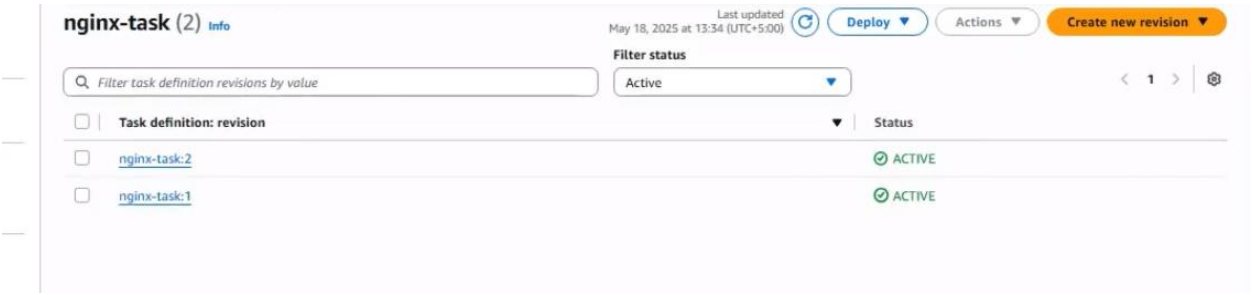
Now add container 2 under Task definition and create new task definition

Select container 2 and keep name like nginx2 and keep image url
Add port mapping and keep 8080 port and keep port name and select http in add protocol
Now create new task definition by clicking button

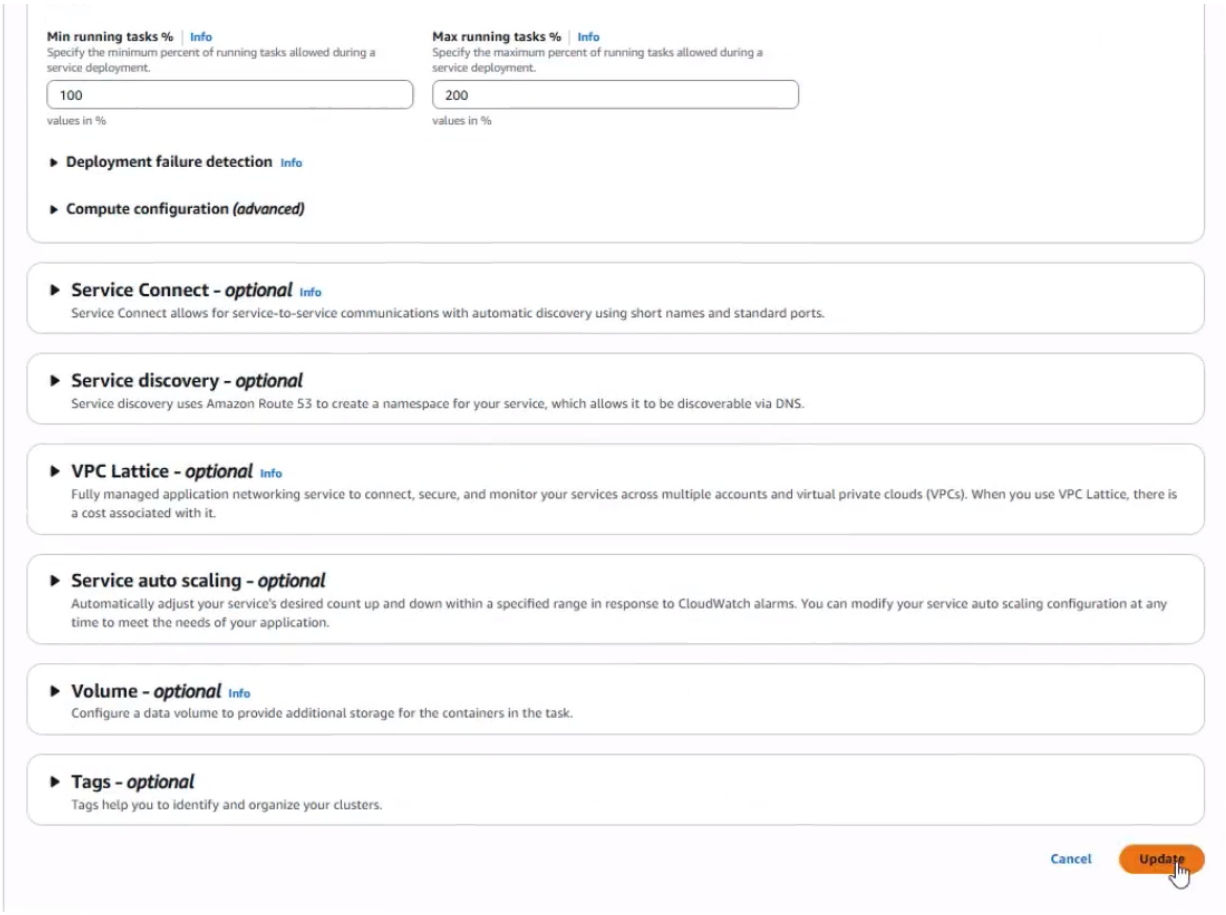


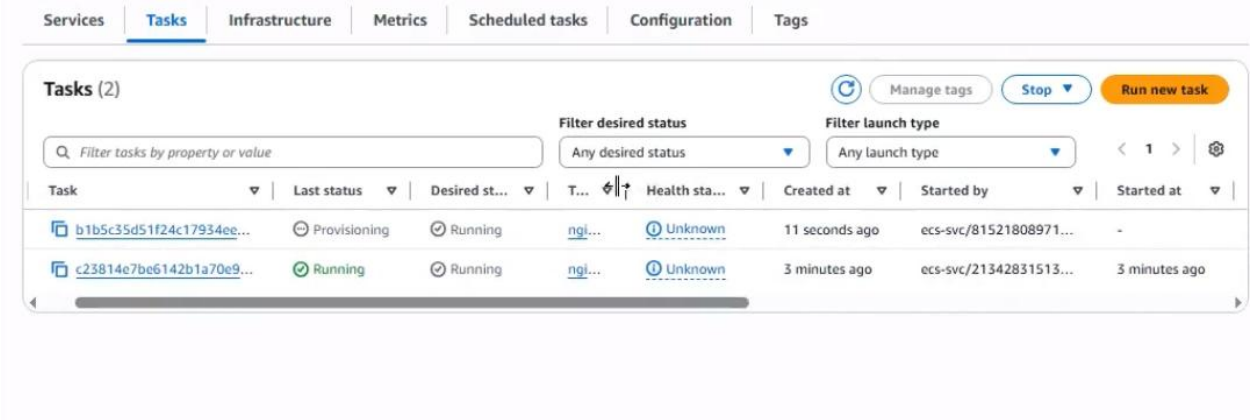

Port mappings | Info
Add port mappings to allow the container to access ports on the host to send or receive traffic. For port name, a default will be assigned if left blank.
Container port
Protocol
Port name
App protocol
8080
TCP
nginx-port-8080
None
Remove
Add port mapping
Read only root file system | Info
When this parameter is turned on, the container is given read-only access to its root file system.
Read only
Resource allocation limits - conditional | Info
Container-level CPU, GPU, and memory limits are different from task-level values. They define how much resources are allocated for the container. If container attempts to exceed the memory specified in hard limi
container is terminated.
CPU
GPU
Memory hard limit
Memory soft limit
1
1
3
1
in vCPU
in GB
in GB
Environment variables - optional
Environment variables | Info
Add individually
Add a key-value pair to specify an environment variable.
Add environment variable
Add from file
Add environment variables in bulk by providing an environment file hosted on Amazon S3.
Add environment file
You can add 10 more environment files.
Logging - optional
CPU and memory allocation for a sidecar
There are logging options that will automatically add a sidecar to your task definition if it does not already exist. AWS provides CPU and memory adjustment recommendations based on the selected options.


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Task definition successfully created
nginx-task:2 has been successfully created. You can use this task definition to deploy a service or run a task.
View task definition
nginx-task:2
Deploy
Actions
Create new revision
Overview Info
ARN
arn:aws:ecs:us-east-1:25349075625 6:task-definition/nginx-task:2
Status
ACTIVE
Time created
May 18, 2025 at 13:34 (UTC+5:00)
App environment
Fargate
Task role
-
Task execution role
ecsTaskExecutionRole
Operating system/Architecture
Linux/X86_64
Network mode
awsvpc
Fault injection
Turned off
Containers
JSON
Task placement
Volumes (0)
Requires attributes
Tags
Task size
Task CPU
1,024 units (1 vCPU)
Task CPU maximum allocation for containers
CPU (unit)
0 100 200 300 400 500 600 700 800 900 1000
nginx1 nginx2 Shared task CPU
Task memory
3,072 MiB (3 GB)
Task memory maximum allocation for container memory reservation
Memory (MiB)
0 200 400 600 800 1200 1600 2000 2400 2800
nginx1 nginx2 Shared task memory
Containers Info
Container name
Image
Private registry
Essential
CPU
Memory hard/sof...
GPU

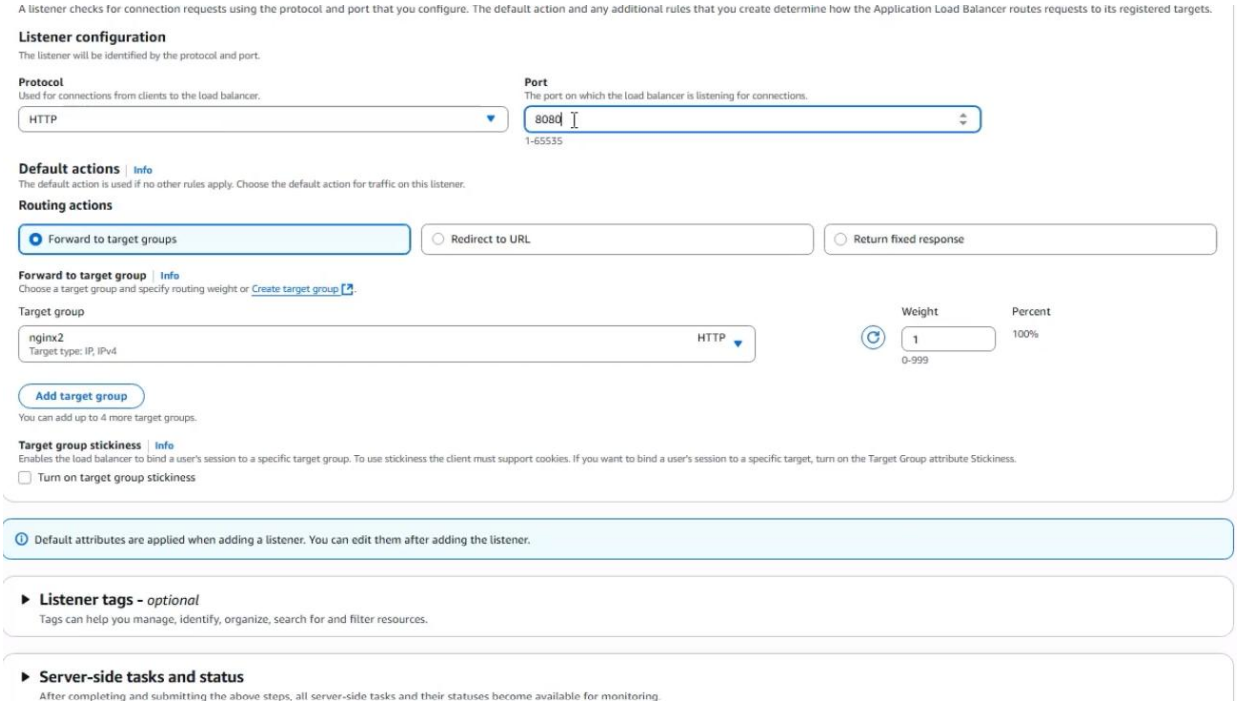


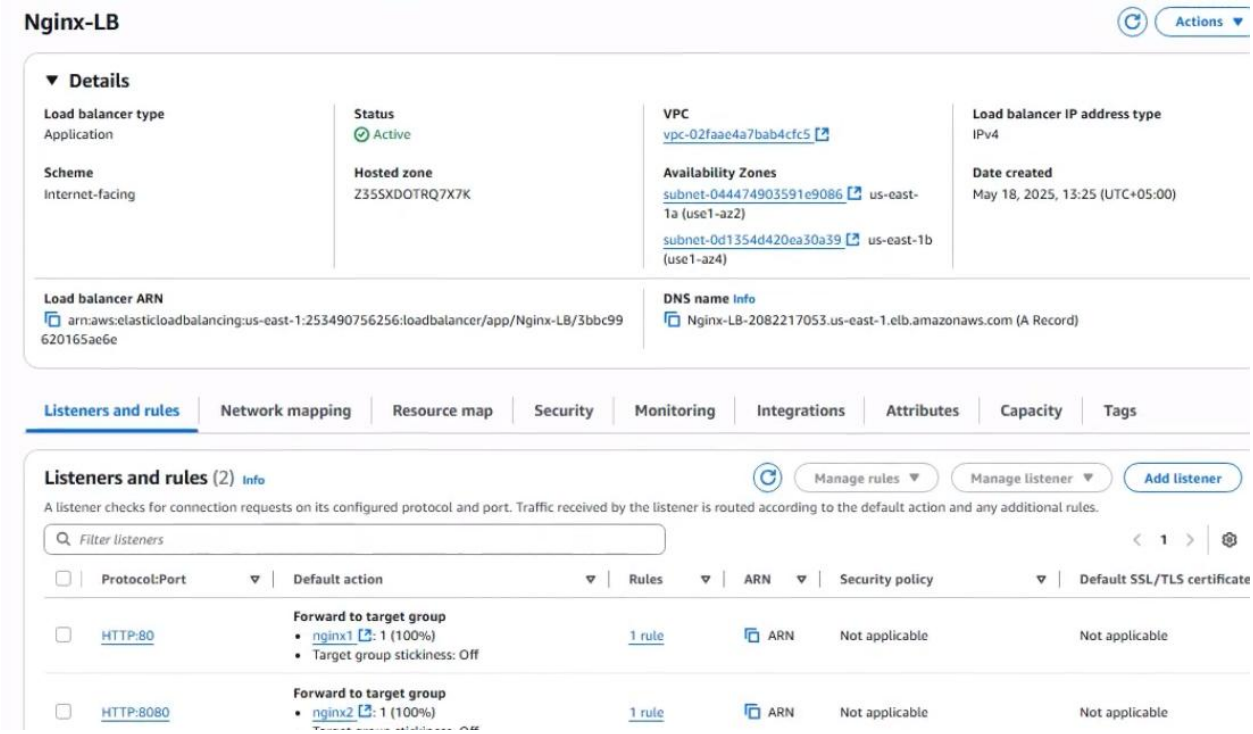
Now update service by changing task definition 2 and then create



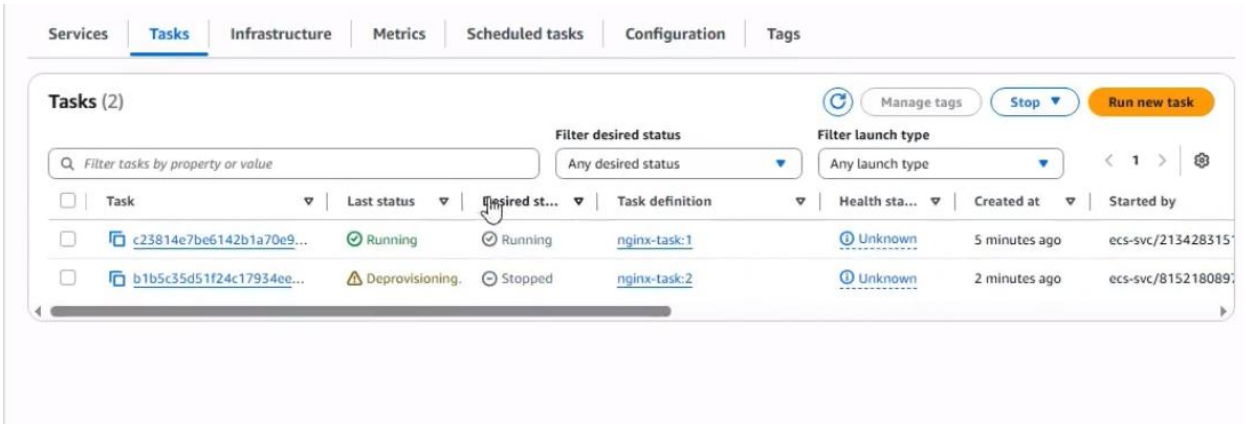


Now go to load balancer
Add listener
Change port 8080
Select nginx2 under Target group

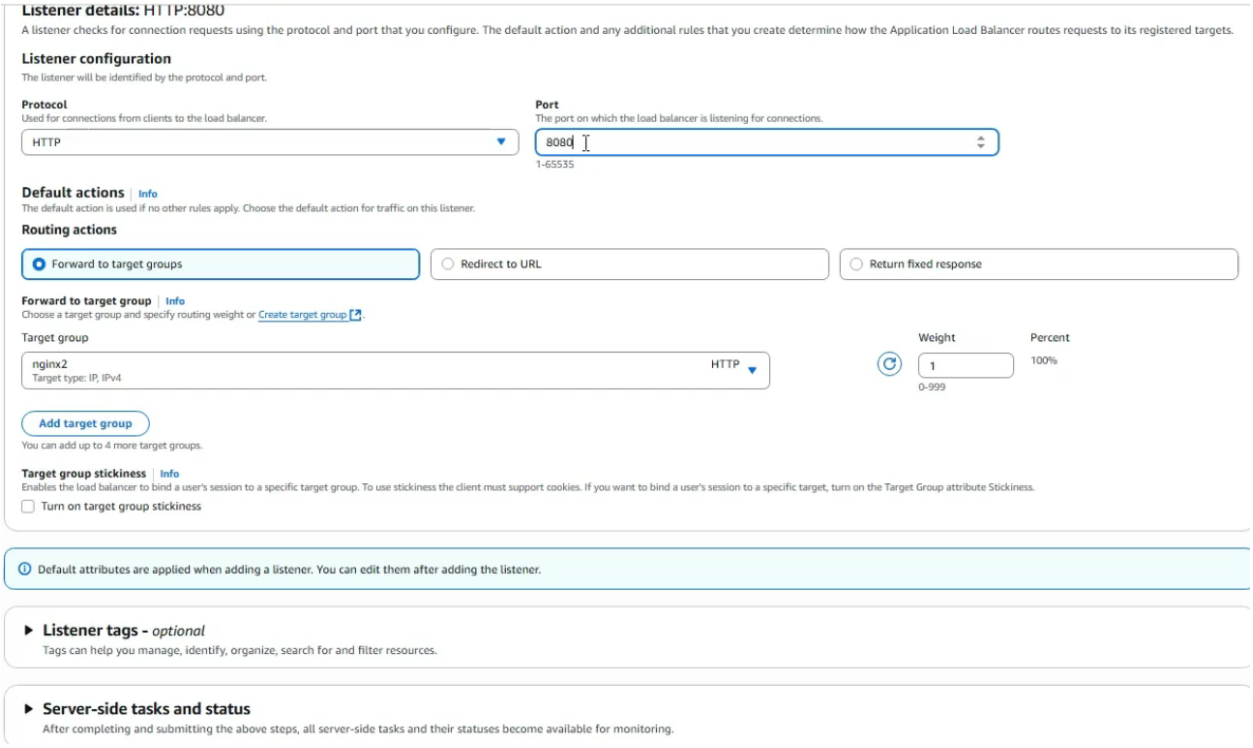




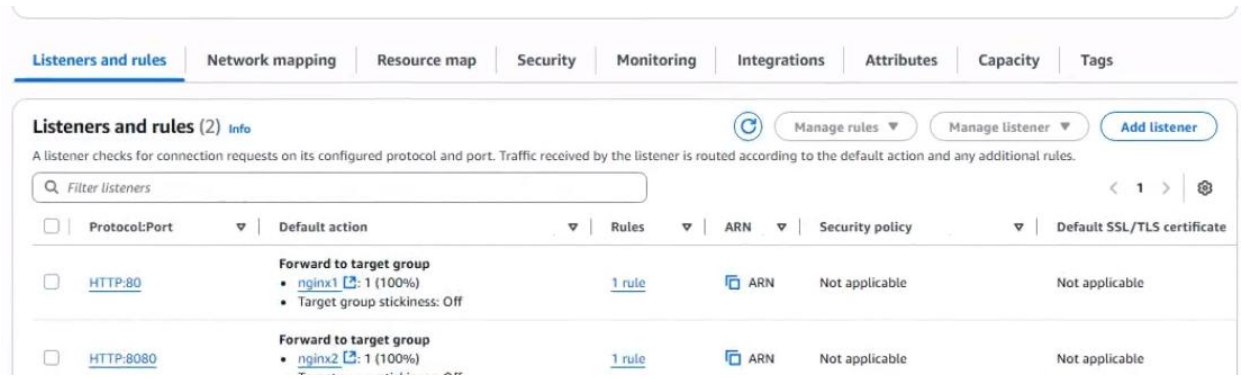
After Editing , you can see the second is deprovisioning



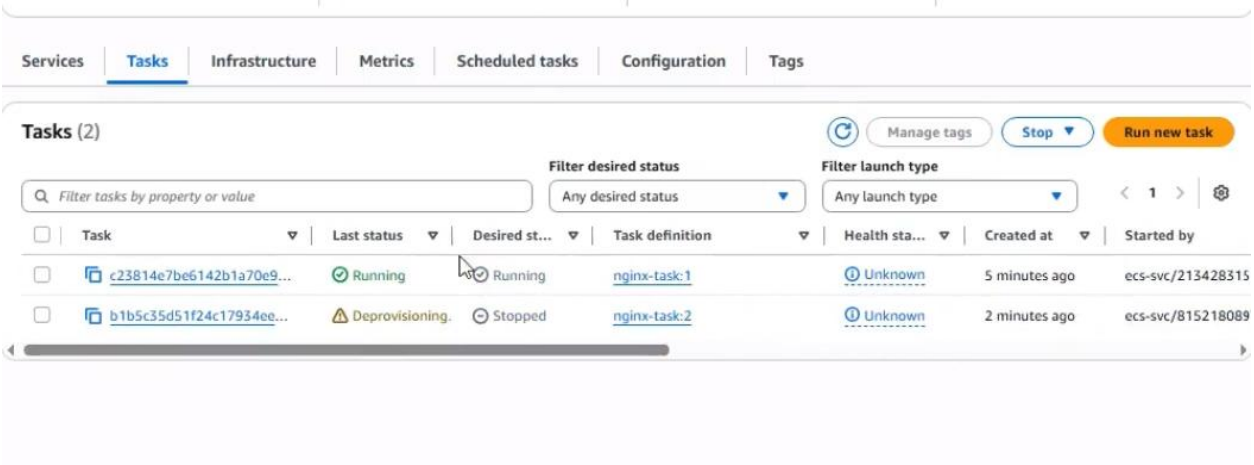
Now go to load balancer and add listener
Select nginx 2
Add port 8080
Then click add button



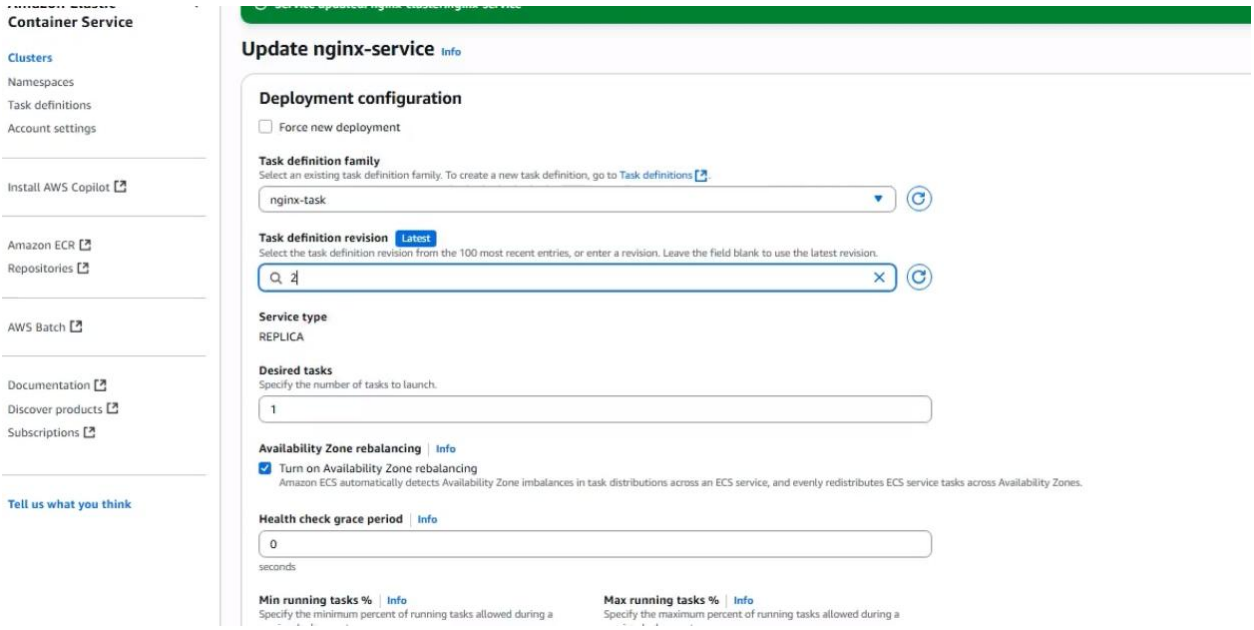
Now we have 2 listeners



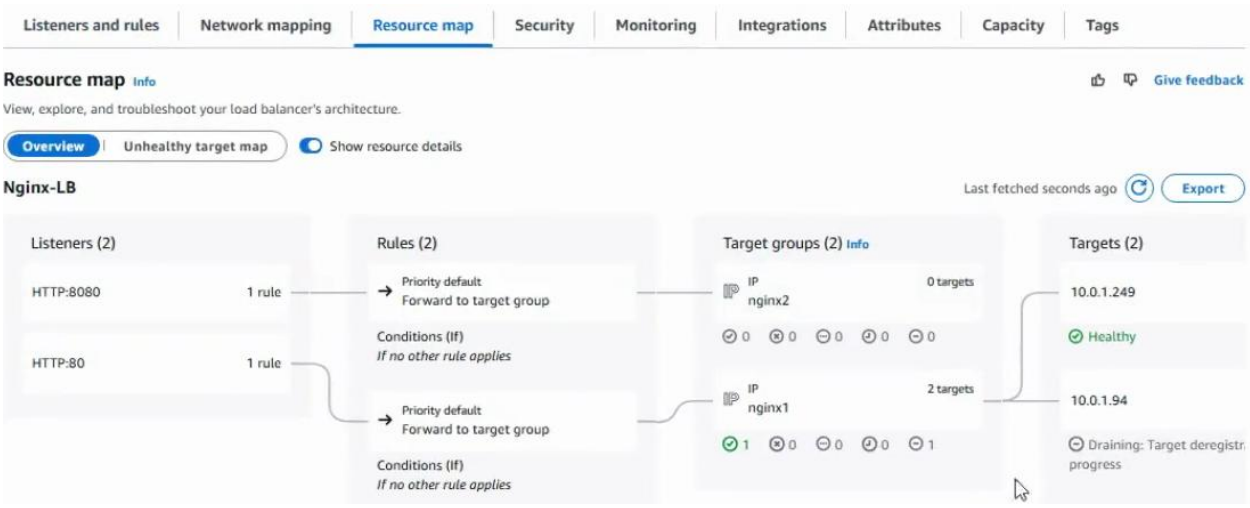
After editing , we have



Now go to cluster and update cluster
Click update service
Select 2 in task definition
And click button



You can see Resouce Map also



Second Output in port 8080:

